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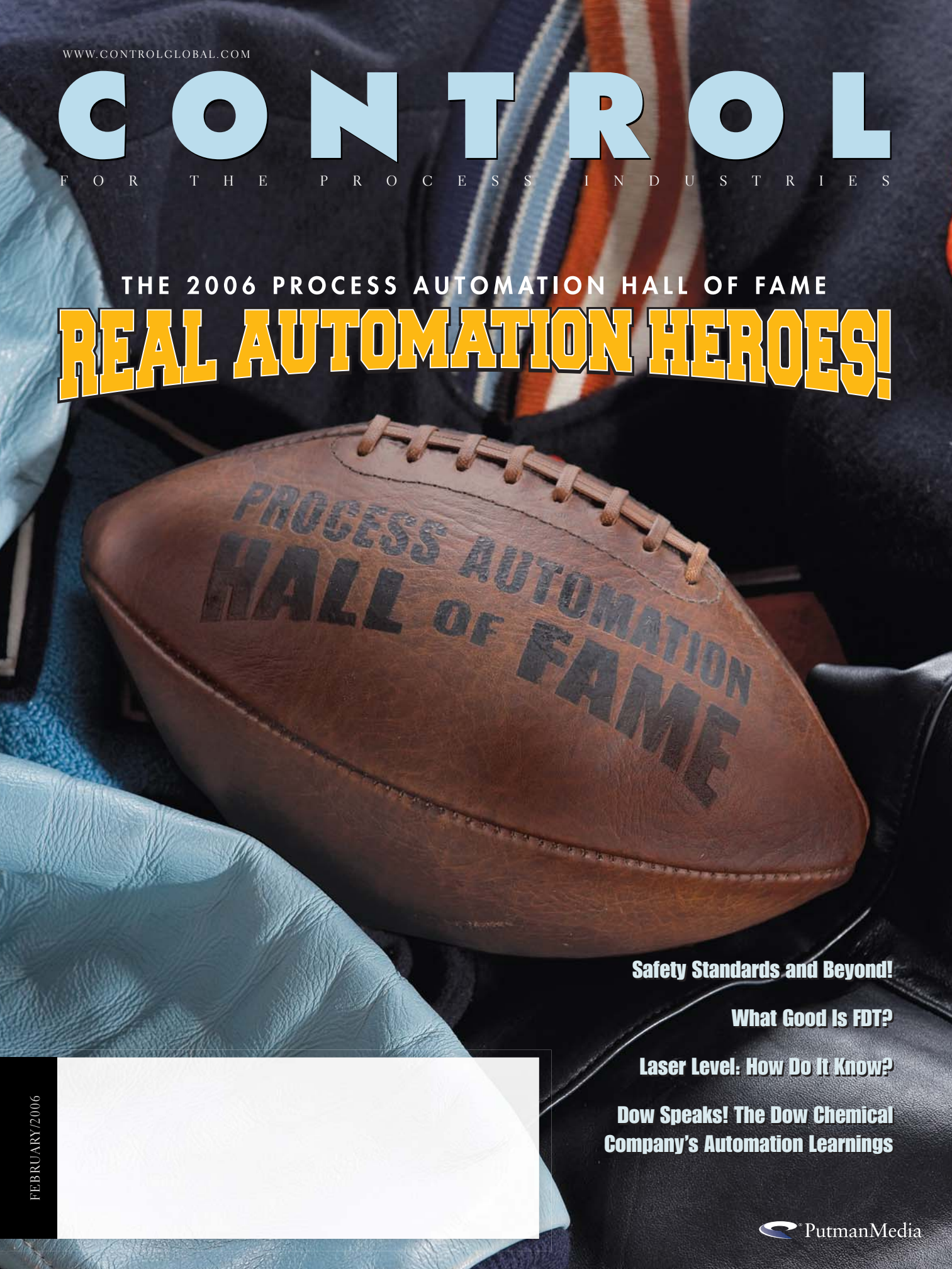

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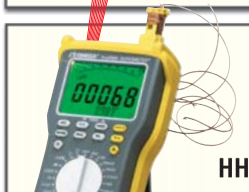

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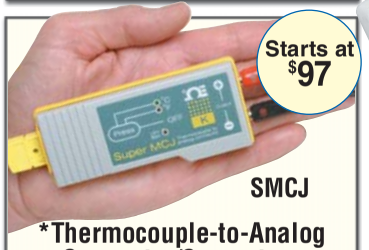

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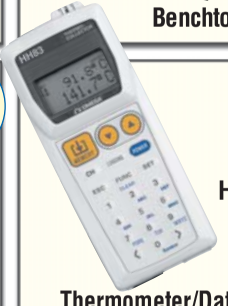

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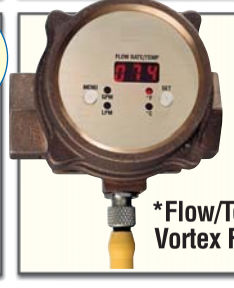

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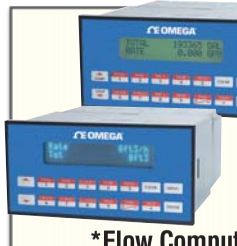

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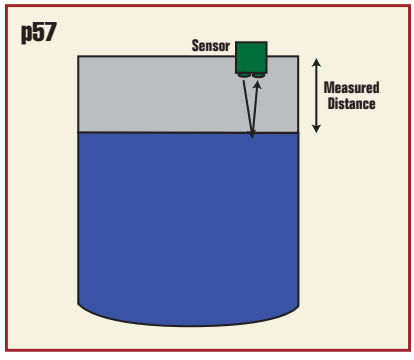

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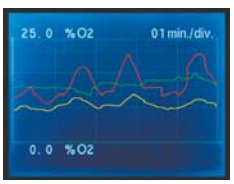

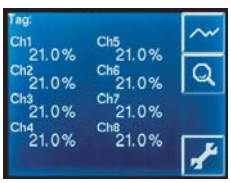

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CIRCULATION AUDITED JUNE/2005

Chemicals & Allied Products	15,926
Food & Kindred Products	15,821
Paper & Allied Products	4,087
Primary Metal Industries	5,341
Electric, Gas & Sanitary Services	2,707
System Integrators & Engineering Design Firms	7,933
Rubber and Miscellaneous Plastic Products	5,016
Stone, Clay, Glass & Concrete products	2,580
Textile Mill Products	2,088
Petroleum Refining & Related Industries	3,361
Tobacco Products	167
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The Models 3100/4100 are pressure/vacuum vents designed to vent the tank vapor away to atmosphere and to relieve vacuum pressure within the tank.

Model 3200/4200

The Models 3200/4200 are pressure/vacuum vents designed to vent the tank vapor away to a header system and to relieve vacuum pressure within the tank.

Model 3300/4300

The Models 3300/4300 are vacuum vents designed to allow atmospheric air into the tank to reduce the vacuum pressure in the tank.

Model 3400/4400

The Models 3400/4400 are pressure vents designed to vent the tank vapor away to atmosphere.

Model 3500/4500

The Models 3500/4500 are pressure vents designed to vent the tank vapor away to a header system.

Model 3600/4600

The Models 3600/4600 are vacuum vents designed to allow atmospheric air into the tank to reduce the vacuum pressure in the tank.

Model 5100

The Model 5100 is a pressure/vacuum vent designed to vent the tank vapor away to atmosphere and to relieve vacuum pressure within the tank.

Model 5200

The Model 5200 is a pressure/vacuum vent designed to vent the tank vapor away to a header system and to relieve vacuum pressure within the tank.

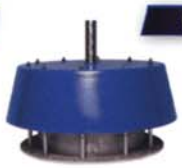
Model 1049

The Model 1049 is a pressure vent designed to vent the tank vapor away to header system similar to the 5500.

Model 1078/1088

The Models 1078/1088 are pilot operated blanketing valves allowing tanks to be controlled with much tighter tolerances than before.

Model 3800



The Model 3800 is a top guided emergency relief vent designed to vent the tank vapor away to atmosphere above the requirements of the standard pressure vent.

Model 3900

The Model 3900 is a hinged emergency relief vent designed to vent the tank vapor away to atmosphere above the requirements of the standard pressure vent.

Model 2100/2200

The Models 2100/2200 are gauge hatches both non-locking and locking respectively for access to tank vessels for gauging and sampling.

Model 2049

The Model 2049 is a tank manifold that allows the end user to utilize a single tank nozzle for both tank blanketing and venting requirements.

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It's For The Future!

If you read the A-list (that's the Automation List, run for years by Control Technology Corp.) for any length of time at all, you notice students writing in for help on projects in automation and control. You also notice that for most of those students, English is strictly a second language.

At a time when a decent automation engineer in India makes about $300 per month, Indian controls students are lining up to take courses like the four-month diplomate in embedded systems design at CDCC-Pune. This course even includes designing wireless embedded systems. The course runs about $1,000, or more than three times an engineer's monthly salary, but CDCC has a waiting list!

Yet here in North America, however, we have a shortage of qualified process automation engineers, technicians and operators.

"We did a study last year (2004)," says a senior engineer, who works for an extremely large oil refining and production company, "of 63 analyzer technicians in [our Gulf Coast] plants, and the average age was 47. Some 'newer' plants had average ages even older, since these folks had been there since the doors opened."

The average age of ISA's members has been increasing with the calendar for nearly 20 years, and the last time I looked it was around 48. When you couple this with the new Process Analyzer Technology (PAT) movement, inspired by the FDA in food and pharma, and Sarbanes-Oxley compliance in other industries, we're not just facing a shortage of trained personnel. We may not be able to actually comply with the government's directives in only a few short years.

For years now, the giant refining and production firm, and its peers, solved the problem by hiring the expertise out. However, analyzer operations and repair companies, and the divisions of large automation companies that also do this work, are swamped, and the pressure is increasing. They have the same problem the refinery does. The average age of analyzer specialists at these companies is, if anything, slightly higher than the average ages of technicians still working at refineries and other plants that use analyzers heavily.

Pretty soon, we're going to either have to import analyzer technicians and engineers in job lots from India and other places, and that may not even be enough, because the economies of India and China are growing fast enough to absorb all of the technically-trained people they can grow.

So we'd better start growing our own.

Companies all over the manufacturing landscape are partnering with schools. I want to highlight one automation company, because they have put their money up front to ensure their future.

It comes from National Instruments' (NI) holiday card: "In the spirit of sharing and in support of our focus on education, NI is reaching out to the Austin Children's Museum to help outfit the museum's after-school and summer robotic camps with Lego Mindstorms for Schools." And NI has even done well by doing good, partnering their LabView software with Lego.

Dear automation vendors: I've shown you what National Instruments is doing as a model for what you should be doing. Please notice that it looks like NI will be doing well, as well as doing good with this project. Not only do they ensure that young people will head toward robotics and automation as career choices by doing this, but they also get to sell product now and in the future.

If you are doing stuff like this, tell me. If you aren't doing stuff like this, tell me why not!

***Walt Boyes,* Editor in Chief**
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While you're here, sign up to receive our exclusive new e-newsletters, covering the latest developments in smart field instruments (DigitalFieldInsider), process control system security (SecureSystemsInsider), and alarm management (AlarmManagementInsider). Subscriptions are free, and if you subscribe, you'll receive a special complimentary white paper by CONTROL's editorial staff, "10 Steps to Avoiding Unnecessary Shutdowns."

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WBF and CONTROL have entered into a content sharing arrangement, and we'll be posting the content archives of the last 12 years of WBF conferences over the next few months. Check back frequently for new batch content.

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When You Blend Canola Oil, Outside Help Improves Workflow.

How a canola oil products maker refined production with Rockwell Automation.

Canbra Foods knows how to spin seed into golden oil. Canbra is Canada's largest, fully integrated oilseed crushing, refining, processing, and packaging company. Their products include cooking and salad oils, margarine, shortening, and a full line of proteins.

Facing stiff competition from soy manufacturers, Canbra sought new ways to increase manufacturing efficiency and reduce costs. So they teamed with Rockwell Automation to control, automate, and monitor production.

A manual-feed solution created a safety solution.

Production begins with canola seed cleaning, then crushing to extract the oil. Finally, the oil is run through a refining process involving a crystallizer machine, making it a finished product for packaging.

In the refining phase, Canbra blends the oil with a custom milk to create margarine products. At this stage, there are eight milk recipes and 10 clean-in-place recipes. That's where production slowed down.

While the oil-preparation process is automated, the milk-preparation stage remained manual and labor-intensive. This created bottlenecks, batching inconsistency, spillage, documentation problems, and a potential safety issue.

Workers faced injury from lifting heavy bags of salt and whey powder, and pouring them into the pre-mixer, on an open floor. Plus, manual operation of the mixer and pumps left room for human error.

Canbra sought to automate ingredient addition in the milk-preparation phase, applying batch management software to reduce waste, increase throughput, and eliminate risk of injury.

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Rockwell Automation automated batching saves Canbra $45,000 a year.

The results were immediate. Automated batching eliminated manual operations, reducing cycle time by half an hour per batch. It also greatly improved batch consistency, made the operation safer by eliminating heavy lifting, and reduced waste from spillage.

The Rockwell Automation batch solution collects data on each batch and compiles it into batch records or event logs, allowing Canbra to identify process trends and deviations, and track ingredients for food-safety audit compliance.

Impressed with current results, Canbra again turned to their partner, Rockwell Automation. The next phase: integrate the oil-blending system with the milk-blending system, to make production flow even smoother. □

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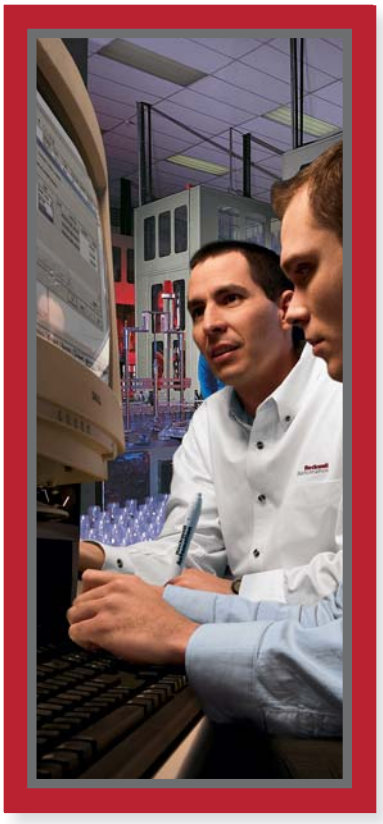

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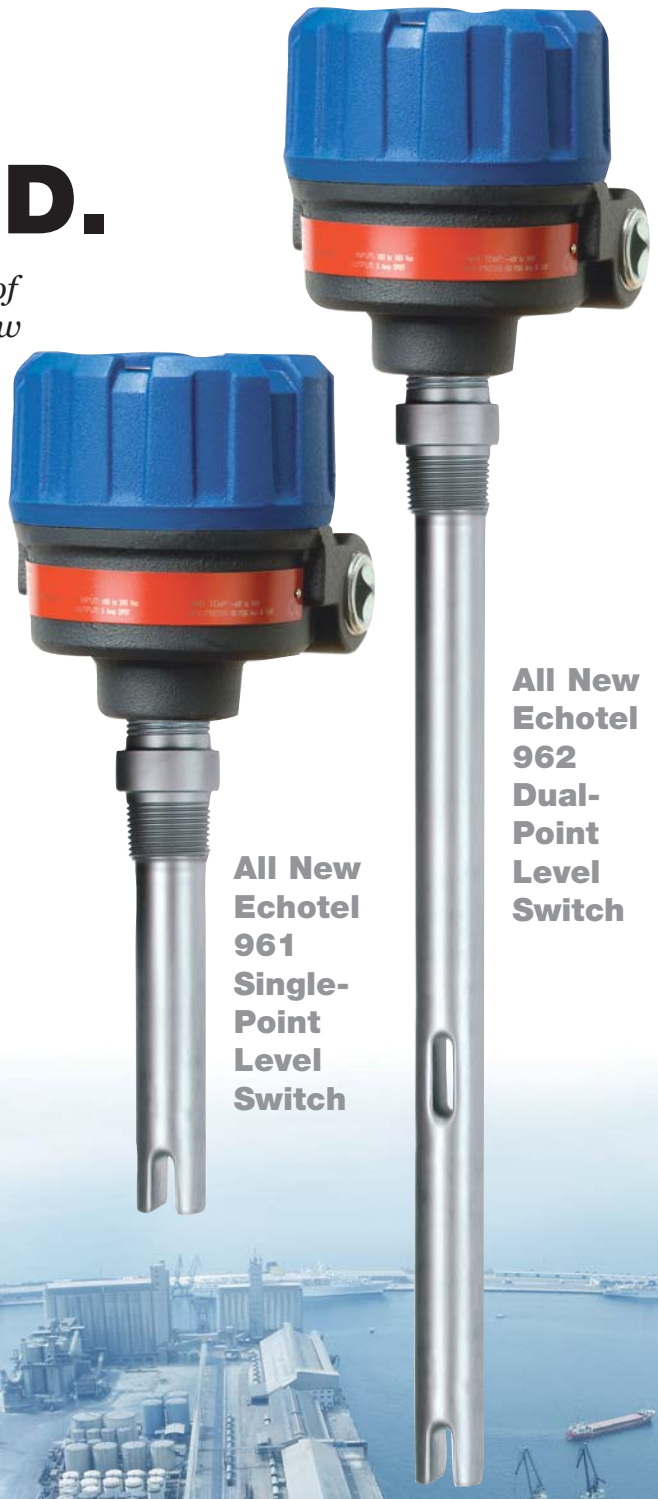

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Global Warming is a Myth!

Okay, the global warming crowd obviously has another robot to spew their misinformation ("A Process Only Mankind Can Control," CONTROL, Jan. '06, p. 20). The myth that mankind can change the weather just by making extremely small changes to the atmosphere is just that—a myth.

You want to see something that can make extreme changes? Check out Mother Nature's own global, weather-changing monster—a volcano. Its power, heat and output dwarf anything we mere specks can do to change the earth's atmosphere. That phenomenon has happened thousands of times since the earth was a smoldering glob of space junk. It formed one of God's greatest creations. A tiny globe, moving through space, carrying and nurturing life-forms. A miracle, but one with cyclical weather cycles.

When the uninformed start carrying around the ideas of an extremely small minority of lunatic fringe "scientists," and preaching it as gospel then people need to see why it's being done. The Birkenstock-wearing crowd has a hard-on for capitalism. They'll do anything to achieve their goal of smashing its ideas. Most of them would rather have us living in some type of socialist/communist state.

The U.S. was built around capitalism. That's how we've persevered for almost 230 years. Immigrants come here in droves to get a small piece of our way of life. They have since our inception. They see us as a way out of oppression, and seek the freedoms we all take for granted. The American dream lives and grows with each of these persons. It's what makes the U.S. the greatest country in the world. I feel very fortunate to have been born, raised, educated, and pursue my career in this country. I strongly feel its time for some publications in this country to start showing common sense. Let us hear both sides about this issue of so-called "global warming."

Brian Walsh
Turbine Diagnostic Services, Tampa, Fla.

Global Warming, Process Control, Hot Air—All Real

In the January issue, my main purpose was to describe the heat-conveying process operating the Gulf Current. The driving force behind this current is reduced as the melting ice-cover of Greenland reduces the density of the water as this stream drops to the bottom of the Atlantic Ocean.

In the March 2006 issue of CONTROL, I'll talk about the atmospheric process of the hurricanes, and in May 2006, I'll cover the exhaustible and inexhaustible energy processes. In each case, I will note the applicable process control theories that can lead to the better understanding of these processes.

Unfortunately, these topics have become political issues, and their discussion tends to result in emotional responses. Yet, I believe that by applying the process control principles to their analysis, such as using artificial neural networks (ANN), our profession not only makes a contribution, but its very existence is recognized.

When I give talks about applying process control approaches to the understanding of such processes as ocean currents or hurricanes, while some might disagree with the ANN conclusions, all are beginning to realize that our new profession does exist, and that it can offer a new tool of understanding from which all mankind can benefit.

Béla Lipták, PE
(former Hungarian freedom fighter)
CONTROL columnist

The Past, Present and Future

Not all that long ago (OK, maybe 40 years), the general state of the art for "controllers" was driven by mechanical relay logic. There were "normally open" and "normally closed" contacts, "latching" relays, "stepping" relays, "thermal time-delay" relays, and many more. Although most of us didn't realize it at the time, those often surprisingly complex control systems were based on the Boolean Logic that became so familiar as we moved to more complex "logic gates." These were composed, first, of individual transistors, then of single-logic gate integrated circuits—ah, those TI 7400 AND, OR, NOR... "DIP" chips (Figure 1)—and finally to today's eminently flexible programmable logic controllers (thanks, Dick Morley!)

Now, again, we sit on the brink of another generation of control logic whose capabilities will be so far beyond that of today's premier PLCs that those devices will eventually feel like the TTL logic chips of older days.

With each technological step, we could do so much more. It took many square inches of relays to implement a simple AND gate. It took perhaps half a square inch to implement a TTL "DIP-chip" gate. Today, tens of millions of logic gates can fit, literally, in less space than your thumbnail. In effect, with each technology generation, we build using ever more complex, individual "Tinker Toys." This lets us incrementally step back, and look at a bigger picture of what we're trying to accomplish, rather than getting bogged down in the details. Today, even simple projects would be a major effort (if realistically possible) using discrete TTL logic chips, and it won't be many years before today's best-of-breed controller capabilities become a similar historical footnote.

Last year, we explored some of the early work that will lead to the next generations of control logic (http://www.controlglobal.com/articles/2004/372.html). However, given the double-exponential growth of many technologies, a year is a very long time, so let's get an update.

As traditional semiconductor manufacturers move to a 65-nanometer process, "...the smallest modern transistors are [now] no more than a handful of molecules across," wrote John Markoff in the Dec. 29, 2005, issue of the *New York Times*. As research marches on, some of last year's speculative ideas are solidifying into devices that may be in our toolkits in as little as nine years; "...The industry's roadmap [has] growing confidence in new technologies that make electronic switches from single molecules or even single electrons."

Indeed, Intel already is planning for 10-nanometer processes that may create chips with a trillion "switches." However, this is seen by the *International Technology Roadmap for Semiconductors, 2005 Edition* as the most optimistic projection for continued scaling of traditional transistors. So here lies a quandary: how can the growth curve of Moore's Law continue as our silicon processes reach fundamental physics limits?

The answer is that, as has always been the case, scientists simply refuse to be held back by any limit. They either find a way of extending the limit, or find a way completely around the limit by thinking outside the box, and then changing all the rules. This is just what they're beginning to do now, as they approach what is commonly seen as a limit to how small conventional transistors can get before they will no longer work.

Unsurprisingly, this is not a smooth road. As experimental transistors shrink toward individual molecules and atoms, the rules of physics that have sustained high school and college physics and all of our manufacturing efforts to date simply refuse to play by the established rules. For example, they defy the traditional belief that a switch must be at a specific, discrete value (such as "on" or "off"), and insist on simultaneously holding values of one, zero and every value in-between! (don't ask; although you can begin to explore this contradiction to common sense at http://en.wikipedia.org/wiki/Quantum_superposition.) Imagine trying to design with something that can represent all values all the time!

Again, as in the past, scientists are working on taming such quantum switches, so that, from the end-user (our) perspective, we simply move to yet another level of Tinker Toy complexity that hides the interior details. This is the same way we evolved from designing with relays to single-gate logic chips to PLCs.

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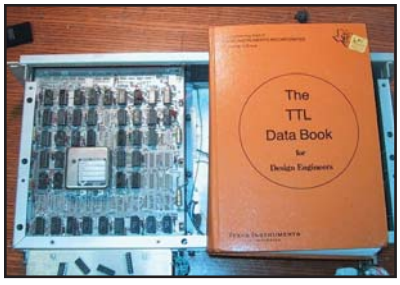
FIGURE 1.

BACK IN THE DAY...



You could only get a small number of switches on a chip.

Still, how can we visualize what this means to us? At this stage of the game it's virtually impossible to imagine the details of what the next generation of Tinker Toys will be able to do, or how we'll combine the new Tinker Toys into functional solutions. Back when we were happily designing products with TI 7400-series chips, imagine how impossible (for most of us, that is) it would have been to foresee the PLC. Now, again, we sit on the brink of another generation of control logic whose capabilities will be so far beyond that of today's premier PLCs that those devices will eventually feel like the TTL logic chips of older days.

We do know that these new devices will be far faster (due to the smaller distances between logic elements on the chip), and will be far more complex (due to the vastly larger number of logic gates that will fit in a given space.) They also will likely be non-volatile, will draw far less power, and generate far less heat, opening up completely new classes of applications. We'll see development of switches that rely on the spin of an electron, on its magnetic state, and on its phase state. And some of our future logic seems likely to blend the realms of the inanimate and the animate, as we hijack nature's biological creations. Already, "single-protein, wet biotransistors" have been demonstrated, and future organic molecular switches hold the potential to switch thousands of time faster than today's. (In-depth insights into additional potential future logic technologies can be found in *Emerging Research Devices*.)

So, as we begin 2006, let's remember that if a project seems too complex or otherwise impossible, most of today's projects would have seemed equally impossible when viewed at the dawn of the PLC. The next generation of control logic might similarly make today's impossible projects quite possible, even easy.

Jeffery R. Harrow is principal technologist with The Harrow Group. He can be reached at jeff@theharrowgroup.com

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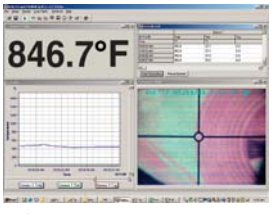

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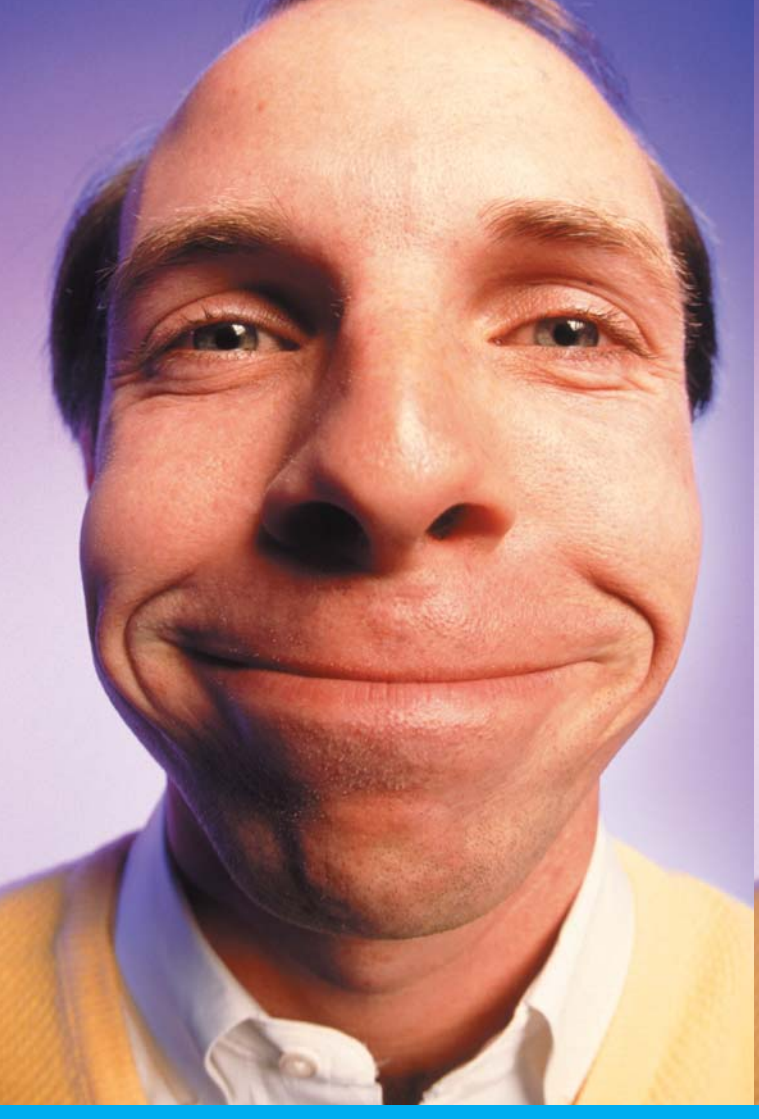

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Tips for Trouble-Free Fieldbus

Though fieldbus projects continue to increase worldwide, many engineers and designers and have yet to use it and reap its benefits. Many believe fieldbus devices aren't much different from what they used in the past, and they're partly right—about the sensor or controller. What's changed is the way that fieldbus devices connect to the rest of the control system, and how they communicate with the host and other network devices.

Depending on installed fieldbus type and the network's configuration, the impact of adding or deleting devices will be different. For example, a Foundation Fieldbus chicken-foot or tree configuration, with multiple field devices terminating in a field junction box and connecting to the host computer's home-run cable, is very similar to what industry traditionally used for many years. Multiple devices connect to a "central" field terminal box, a multi-conductor cable runs to a host computer, and individual signals terminate on an I/O card (Figure 1).

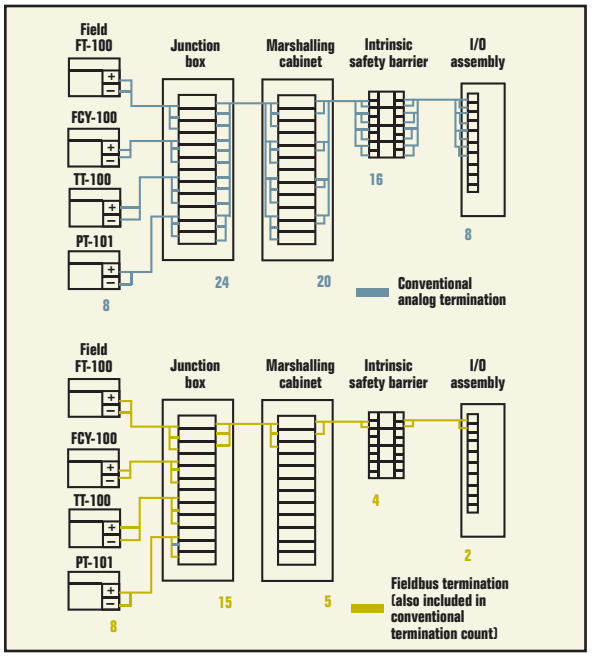
Because a fieldbus is a network, however, it also requires terminators at each end of that network to balance impedance, and avoid reflections and signal degradation. Termination details for each fieldbus network are different, and must be given careful consideration during design. Similarly, calculations on network loading, voltage and current at each device, as well as cycle time, jitter and cable layout, must be considered in the design.

If engineering and design are done properly and documented, then field construction shouldn't be much different than a conventional system. Special tools are required to monitor how the infrastructure supports digital communications, although conventional analog devices can monitor voltages, currents and continuity. Most fieldbuses have simple handheld troubleshooting tools to conduct preliminary analyses.

The most important measure that can be taken to prevent difficulties with a given fieldbus or digital networking project is protocol communication.

FIGURE 1.

SIMILAR LOOKS, KEY DIFFERENCES



Because a fieldbus is a network, it requires terminators at each end to balance impedance, and avoid reflections and signal degradation.

Further, operations and maintenance staff must understand how the control system will handle alarms. In the early days of Foundation Fieldbus, for example, one control system supplier couldn't distinguish between an alarm from a failed device on the network and an alarm caused by a failed system controller. Both were treated as system-level alarms, which they were, since both occurred on the network and, hence, at system level. But, due to the alarm's severity, it wasn't possible to disable the device from alarming, and so any other system-level alarms would have been masked by this one because a flow transmitter had failed.

However, the operators thought it was the fieldbus that didn't work because they always got this alarm. In fact, it was working as it was supposed to by annunciating the device failure.

As the above incident with the flow transmitter shows, the most important measure that can be taken to prevent difficulties with a given fieldbus project is communication.

Ian Verhappen, contributing editor
Ian.Verhappen@ICE-Pros.com

OSHA Endorses Three-Part ISA-84 Standards

Will help comply with rules on preventing and mitigating accidents involving hazardous chemicals

The U.S. Occupational Safety and Health Administration (OSHA) has notified ISA (www.isa.org) that its three-part ISA series of standards may be used to help comply with OSHA's regulations on prevention and mitigation of accidents involving hazardous chemicals. These three standards are ANSI/ISA-84.00.01-2004 Parts 1-3 (IEC 61511 Modified), "Functional Safety: Safety Instrumented Systems for the Process Industry Sector." The standards resulted from the adoption and modification of the three-part international standard IEC 61511 by ISA's SP84 standards development committee. Leaders of ISA-SP84 helped develop IEC 61511.

OSHA's rule to which the standards may be applied is 29 Code of Federal Regulations (CFR) 1910.119, "Process Safety Management (PSM) of Highly Hazardous Chemicals." It establishes the need for PSM procedures to protect employees by preventing or minimizing chemical accidents involving highly hazardous chemicals. OSHA's endorsement of ANSI/ISA-84.00.01-2004 Parts 1-3 reportedly is vital in enabling companies to use the standards to help meet OSHA requirements for employee safety.

"OSHA considers ANSI/ISA-84.00.01-2004 Parts 1-3 (IEC 61511 Mod) to be recognized and generally accepted good engineering practice for safety instrumented systems," stated Richard Fairfax, director of OSHA's Enforcement Programs Directorate, in a letter to ISA. "Therefore, if an employer chooses to use ISA-84.00.01-2004 Parts 1-3 as a basis ('code or standard employed') for SIS, and meets all ISA-84.00.01-2004 Parts 1-3 requirements and other OHSA PSM requirements related to SIS, the employer will then be considered in compliance with OSHA PSM requirements for SIS."

SP84's chairman, William Johnson of E.I. DuPont, says, "Endorsement of ANSI/ISA-84.00.01-2004 Parts 1-3 by OSHA as a generally accepted good engineering practice for safety instrumented systems (SIS) is a major step in reinforcing IEC 61511 as a multinational standard. This will have a global impact on the functional safety of process plants. The challenge now is to accomplish the goal of increasing functional safety and reducing the lifecycle costs of SIS designed to the new standards."

To help meet that challenge, ISA published two major technical reports in January 2006 that were developed by ISA-SP84 for use in conjunction with the standards:

- ISA-TR84.00.04-2005 Part 1: Guideline on the Implementation of ANSI/ISA-84.00.01-2004 Parts 1-3 (IEC 61511 Mod)
- ISA-TR84.00.04-2005 Part 2: Example Implementation of ANSI/ISA-84.00.01-2004 Parts 1-3 (IEC 61511 Mod)

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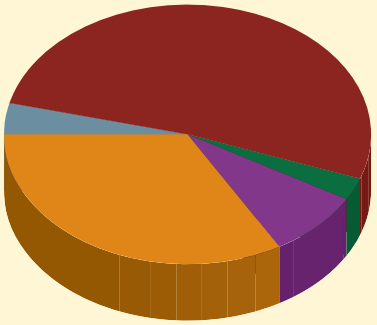
Do you feel you're up-to-date on the skills you need to do your job?



- 4% Heck yes! I passed ISA's new CAP exam without studying
- 51% Mostly, but it is a struggle to keep up
- 3% Sometimes, but most of our staff is older, and if we don't upgrade, I'm fine
- 8% I'm so new that everything I know is current
- 33% I gave up trying long ago. Now, I just put out fires

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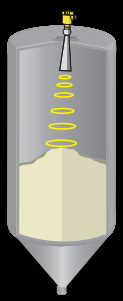

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ABB Automation World Conference & Exhibition

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ABB's Automation World was started to serve automation industry end users and specifiers who were looking for ways to improve operational performance. It has grown to be one of the largest Automation supplier hosted events, held in North America, by delivering what industry experts like you demand... ***results!***

Many aspects of the event are reviewed and approved by the Customer Advisory Board. This group of end-users reflects the diverse automation marketplace and has strong opinions about what attendees, like yourself, need from an event like this. In cooperation with the Customer Advisory Board, we make sure that ABB's Automation World is designed to give you real ideas and solutions that you can take back and use immediately.

The Customer Advisory Board helps to ensure that the event provides you with the following benefits:

- An event for various job functions and industries. As operations become more integrated, thanks to technology like Industrial IT, it is increasingly important for all job functions to understand how to leverage the data that modern automation solutions provide.
- An opportunity to meet and socialize with your peers and industry experts to learn from each other and discuss issues, problems and challenges. In fact when you register, you have the opportunity to ask for specific topics to be addressed at the conference.
- A chance to see what the future holds in power and automation technology. ABB and industry experts give you insight into how future technology will bring advantages to your business.

Countdown to Results

ABB Automation World Conference & Exhibition

"It gets better every time."
Vincent Chiew
City of Calgary

"I was very much impressed with the organization of the whole conference."
Stephen Sliwinski
Gramercy Alumina LLC

"It was a very classy presentation."
David Summers
NPA Automotive

"My experiences from the past two year's events were all good."
Kien Van Vo
DuPont Fluoroproducts

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Countdown to RESULTS

Over 40,000 square feet of demonstrations

The exhibition area includes demonstrations of ABB's Industrial IT architecture. Also featured are ABB's systems, software, products and services, along with displays by select third-party technology partners... making it one of the most comprehensive technology solutions showcases ever held in one place.

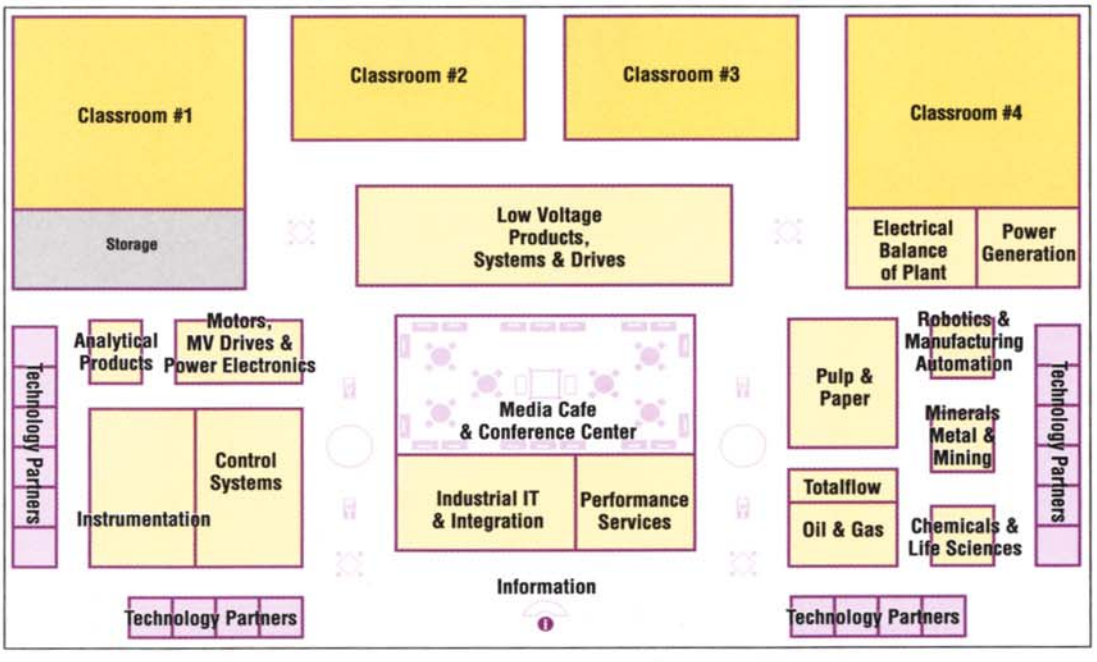
"Truly innovative companies generate a major segment of their revenue from products designed within the past three to five years. ABB... stands out in this respect. ABB allocates more than 8% of its revenues and 20% of its corporate resources to new developments and improvements."

Jim Pinto's eNews - December 2005

"The recent ABB Automation World bears out the fact that ABB is back."

Control Magazine Editor Walt Boye's Weblog "Soundoff" - April 2005

Value added solutions featured in the demonstration area include Industrial IT System 800xA, open control system evolution, safety instrumented systems and high integrity control applications, industry-specific solutions for chemical, life sciences, metals & mining, pulp & paper, oil & gas manufacturing, electrical utilities, water and wastewater.



NEW for 2006 Automation World:

- 40,000 square feet of exhibition space
- Solutions desk
- Media cafe
- Executive track
- Integrated exhibits
- Panel discussion with executives
- Technology partner displays
- Future view of technology
- Evening training
- Free training throughout event

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Workshops, training and customer case studies

Over 200 workshops, training and customer case studies have been carefully developed to meet the needs of all attendees. With more customer-led workshops in 2006, this is a great opportunity to learn best practices, successes, tips and proven methods that will make your job easier and improve your overall business performance. In

2006 Automation World Conference and Exhibition

May 9, 10, 11

Hilton Americas Houston, TX

Paid conference program includes:

- General sessions
- Customer case studies
- Exhibit hall
- Application oriented workshops
- Technical training
- On-site meals and off-site events
- Executive track
- FREE iPod for registering by April 3

Improving business operations

- Increased business visibility through integrated information management
- ROI from your automation investments
- Improving hidden plant equipment effectiveness through real-time production intelligence

Control system integration, application and evolution

- Enhancing existing control systems with System 800xA
- System lifecycle management
- Improving time from decision to action

Asset optimization and device management

- Optimizing plant asset availability
- Reliability centered maintenance strategy
- FOUNDATION Fieldbus, HART and PROFIBUS *

Safety, security and alarm management

- The do's and don'ts of SIL determination
- Safety regulations IEC61508 and IEC61511
- Alarm management

Chemical industry applications

- Safety systems
- Composite plant solutions
- Analytical technology solutions

Life sciences industry applications

- Streamline validation
- Batch profitability, consistency & traceability
- Meeting cGMP needs with System 800xA

Oil and gas industry applications

- Asset management for oil and gas production
- Gasflows and Totalflow
- ATEX motors in hazardous areas

Metals and mining industry applications

- Proactive control and optimization in a cement plant
- Shovel and dragline retrofits
- Integrated composite plant and electrification package

Pulp and paper industry applications

- FEED studies for paper OCS evolution projects
- New low and medium voltage drives
- Paper machine optimization services

Water and wastewater applications

- Condition monitoring
- Reliability centered maintenance strategy
- Insitu flowmeter calibration verification and condition monitoring

Industrial steam automation applications

- System 800xA for Harmony/INFI 90
- Substation automation utilizing numerical protection and bay control
- RTU-based substation automation solutions

Utility automation applications

- Increased functionality through console migration
- EBOP integration
- Combustion optimization

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"I really enjoyed it."

William Carter
The Dow Chemical Company

"I felt that I gained a wealth of knowledge and I was genuinely impressed with how ABB was able to handle so many people and different topics. The event went smoother than any that I have experienced in the past. I am looking forward to participating in 2006."

David Kavanaugh
Bayer Corporation

"I felt that the conference was extremely well done and I thoroughly enjoyed all the events!"

George Sydnor
NASA

FREE iPod Shuffle! Register online by April 3*

*See website for full details. iPod Shuffle contains conference agenda and information. Paid conference attendees only.

... all designed to deliver *results*!

addition, all technical workshops will offer a "certificate of completion" so that you have a record of the training you will receive during the event. Below is an example of some of the topics to be covered during the three days of the event.

Electrical systems specialist applications

- Bundling and packaging electrical balance of plant
- Protecting personnel and reducing "arc flash exposure"
- Transformer fleet management

Motors, drives and power electronics specialist applications

- Drives and motor sizing made easy
- Now my motor is a generator!
- What is a fuel cell and how does it work?

Analyzer specialist applications

- pH sensor technology theory, installation and verification
- Vista II GC analyzer set-up and metering
- ABB analytics assured performance agreements

Field device specialist applications

- Pressure transmitters and remote seals application and selection
- Temperature measurement primer
- Damper control actuators

Flow measurement specialist applications

- Flow technology
- Flow measurement in extreme applications
- Groundwater discharge monitoring

Robotic industry applications

- Pick, pack, palletize
- Oil industry welding successes
- Robotic applications

2006 Automation World Agenda-at-a-Glance

	Monday May 8	Tuesday May 9	Wednesday May 10	Thursday May 11
AM 7		Breakfast	Breakfast	Breakfast
8		General Session ABB & Customer Keynote Speakers	Workshop Session 5	General Session
9		Break	Break	Break
10		Workshop Session 1	Workshop Session 6	Workshop Session 11
11		Lunch	Break	Break
Noon 12			Workshop Session 7	Workshop Session 12
1		Workshop Session 2	Lunch	Lunch
2		Break	Workshop Session 8	Workshop Session 13
3		Workshop Session 3	Break	Break
4		Break	Workshop Session 9	Workshop Session 14
5		Workshop Session 4	Break	Break
PM 6		Workshops 5:30-8:00 PM Exhibits Open Until 9:00 PM	Workshop Session 10	Workshop Session 15
7	Welcome Reception & Dinner with Guest Speaker		Offsite Venue and Dinner Johnson Space Center	Reception and Exhibits Open Until 7:00 PM
8				

Be sure to arrive by 6:30 p.m. Monday, May 8th to attend the Automation World Welcome Reception and Dinner with Guest Speaker!

"My first event, not my last, I truly enjoyed it."
Dan Neuts
Hiram Walker & Sons

"I learned a lot and it was very informative."
Florian Reid
General Chemical Corporation

"I thought the meeting was very good and well organized."
Leslie Davis
Boise Paper

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For the seminars, scroll down to the Web Seminars section and select "Browse and select your free web seminars." For the papers, scroll down to the Technical Papers section and select "Members (full dues) access the full library."

NMW Gets New Home, Bigger Conference

Don't count National Manufacturing Week (NMW) out just yet. Sure, the venerable, multi-show event has shrunk due to the Internet's emergence, possible over-diversification, and the growth of niche-based tradeshows and corporate preach-to-the-choir events. However, organizers say they've signed up more than 1,000 exhibitors for this year, staged 12 industry-specific pavilions, put together an aggressive 250-session conference, and scheduled seven free keynote presentations.

NMW 2006 (www.manufacturingweek.com) will offer its usual four-show format on March 20-23 at the 840,000-sq-ft Donald E. Stevens Convention Center in Rosemont, IL, just outside of Chicago and next to O'Hare International Airport. The four main shows are Design Engineering, Industrial Automation, Plant Engineering and Enterprise IT.

The conference will be held:

- 9 a.m. to 5 p.m. on March 20-21,
- 9 a.m. to 4 p.m. on March 22, and
- 9 a.m. to 12 noon on March 23.

The show floor will be open:

- 10 a.m. to 5 p.m. on March 21-22, and
- 10 a.m. to 3 p.m. on March 23.

Organizers add that NMW's expanded conference will focus on the needs of small and mid-sized manufacturers. The conference will contain five tracks, 23 sub-tracks, and feature more than 400 speakers. The eight keynotes will include:

- Jack Perkowski, ASIMCO's chairman and CEO, who will present "China: Opportunities and Challenges." Perkowski was featured in the recent best-seller, *The World is Flat*, by *New York Times* columnist Thomas Friedman
- Jim Owens, Caterpillar's chairman and CEO, will present "Manufacturing Leadership in the Global Marketplace."
- Colin Wu, China Business Sources' president, will present "What's Behind Chinese Competitiveness?"
- Doug Engel, Deloitte and Touche USA's vice chairman, and Tommy Thompson, Deloitte Center for Health Solutions' chairman, will deliver "Managing Structural Costs and Maintaining Competitive Edge: Improving Your Position in the Global Marketplace."
- Gilles Bouchard, Hewlett Parkard's global operations EVP, will deliver "Customer Choice: Managing Multiple Routes-to-Market on a Global Scale."
- Mike Santori, National Instruments' automation fellow, will present "How to Compete in a Global Economy."
- Mary Frances Cox, Schneider Electric's senior operations VP for North America, will present "Achieving significant reductions in manufacturing cost base."

Hydraulic Institute Launches Viscosity Correction Standard

The Hydraulic Institute (HI, www.pumps.org) has published its new "American National Standard for Effects of Liquid Viscosity on Rotodynamic (Centrifugal and Vertical) Pump Performance (ANSI/HI 9.6.7)." This standard replaces HI's more than 40-year-old viscosity correction nomographs, and is designed to predict performance changes of rotodynamic pumps operating on viscous liquids exhibiting Newtonian behavior relative to their performance on water, which reportedly is the basis for most published curves.

"By eliminating the need for interpolating graphic solutions, this standard has major technical significance and practical applications," says Gregg Romanyshyn, HI's technical director. "This landmark standard will provide valuable guidance to pump manufacturers, users, and others who have an interest in the performance of rotodynamic pumps."

HI members developed the new standard along with representatives from France, Germany, Italy, Japan, and Switzerland. It also includes contributions from industry experts and veterans.

The standard covers conventional rotodynamic pumps, including open or closed impellers and single or double suction pumps. It further contains information on methodology, definitions of symbols and terms, flowcharts that outline the procedure, theoretical explanations behind the methodology, and a method for estimating net positive suction head required (NPSHR).

PLCopen Seeks Comments on TC3 TF PLC Benchmarking Standard

PLCopen Task Force is seeking comments on its recently proposed TC3 TF technical specification, which has been released for comments until June 30, 2006. The organization is seeking feedback from a broader community about the specification's usability for their environments before developing the applicable test software. The published version is designated as 0.4, and is combined with a feedback form.

Though a benchmark is a reproducible, portable test to measure the performance of a given system in comparison to other systems, the task force reports there are no defined benchmarks for PLC systems. The only common measurement for PLC performance is the execution speed of 1,000 Boolean operations, but this isn't a good benchmark because users can't derive PLC performance in a typical application from it. Consequently, it's not comparable between systems because the test conditions lack definition.

The group adds there are two main objectives when using a benchmark: to estimate the performance of the PLC in your own application; and to compare the performance of the PLC with other PLCs, and find out the specific strength and weakness of a given system

PLCopen's paper defines two different sets of benchmarks to meet the objectives described above. The first set of benchmarks defines five different types of applications, which are typical for using a PLC. Most applications belong to either to one of these types or consist of a mixture of them. The second set of benchmarks measures each language feature of the IEC 61131-3 separately. The goal of this test is to exclude effects of other features as much as possible.

The released document is available free-of-charge at PLCopen's www.plcopen.org website under "TC3–Certification/Task Force Benchmarking."

Chinese End Users to Highlight FF General Assembly

The Fieldbus Foundation reports the keynote speakers for its 2006 General Assembly, Feb. 28-March 1, at the Ritz Carlton Portman Hotel in Shanghai, China, will include Huang BuYu, SINOPEC Engineering's deputy chief engineer, and Seah Ooi Kiat, CNOOC and Shell Petrochemicals' head instrument engneers. They will describe how Foundation Fieldbus technology allows implementation of modern, open and interoperable plant automation strategies, and enables improved productivity and profitability in global industrial markets.

The foundation's general assembly also will address the needs of the expanding Foundation Fieldbus market in China and Asia/Pacific, and will include case-study presentations by leading end users from throughout the region. It will also feature a series of tutorial workshops by leading fieldbus technology experts. Throughout the event, foundation-member companies will present tabletop exhibits displaying their latest Foundation Fieldbus devices, tools and host systems. For more information or to register, visit www.fieldbus.org.

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People for Process Automation

Emerson Re-Issues Pressure Gauge Manual

Emerson Process Management has published the fifth edition of its *Pressure Regulator Application Guides*. Previously combined into one volume, known as the *Regulator Handbook*, these guide now are available in two volumes, *Natural Gas Regulators* and *Industrial Regulators*.

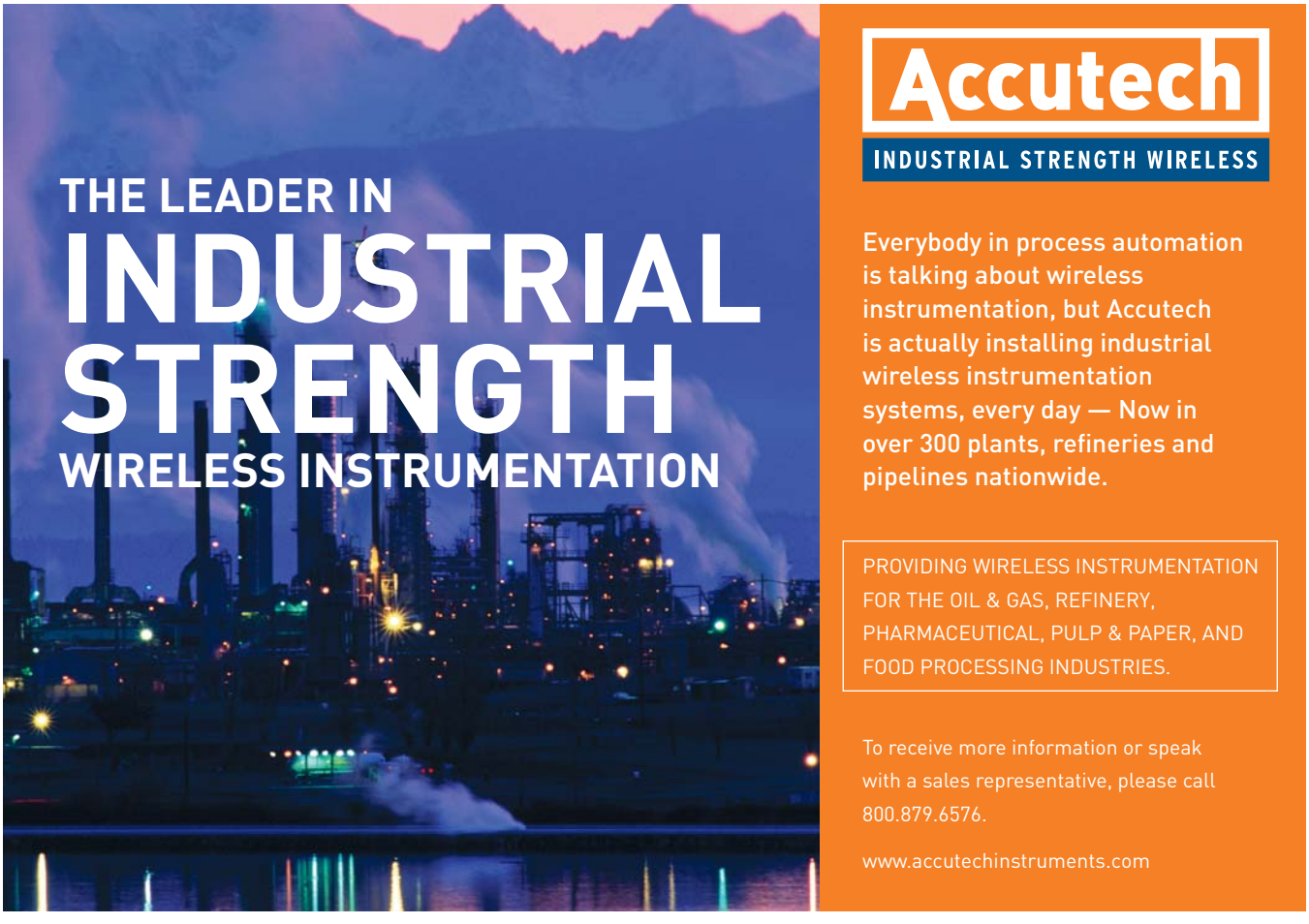
The guides are a reference for technical applications and pressure data on Emerson's Fisher natural gas and industrial regulators, and are designed to provide assistance in selecting the right pressure regulator for a broad range of applications anywhere. The guides provide detailed information on the principles of various products and applications such as air, steam, tank blanketing, liquids, and natural and process gases.

The 465-page *Industrial Application Guide* has the same technical information as the previous handbook along with new information on Fisher sanitary regulators. Similarly, 580-page *Natural Gas Application Guide* contains all previous handbook data, along with information on all new Fisher regulators. The guides are useful for engineers, maintenance technicians and supervisors, and those in purchasing or production/plant operations. For more information, visit www.fisherregulators.com

Pocket Foundation Guide Published

The second edition of *Foundation Fieldbus: A Pocket Guide* by Ian Verhappen and Augusto Pereira has been published by ISA in an easier to read format. In this edition, the authors summarize Foundation Fieldbus technology, show how it works, and provide tips and useful tools. The book also contains handy forms to help with project design, commissioning, and a new section on operations and maintenance.

Also new to the second edition are a discussion of Fieldbus Safety, High Speed Ethernet, FISCO, FNICO and the enhancements being made to Electronic Device Description Language (EDDL). The appendices include useful reference information on segment design tools, as well as a comprehensive listing of the acronyms used to describe Foundation Fieldbus.



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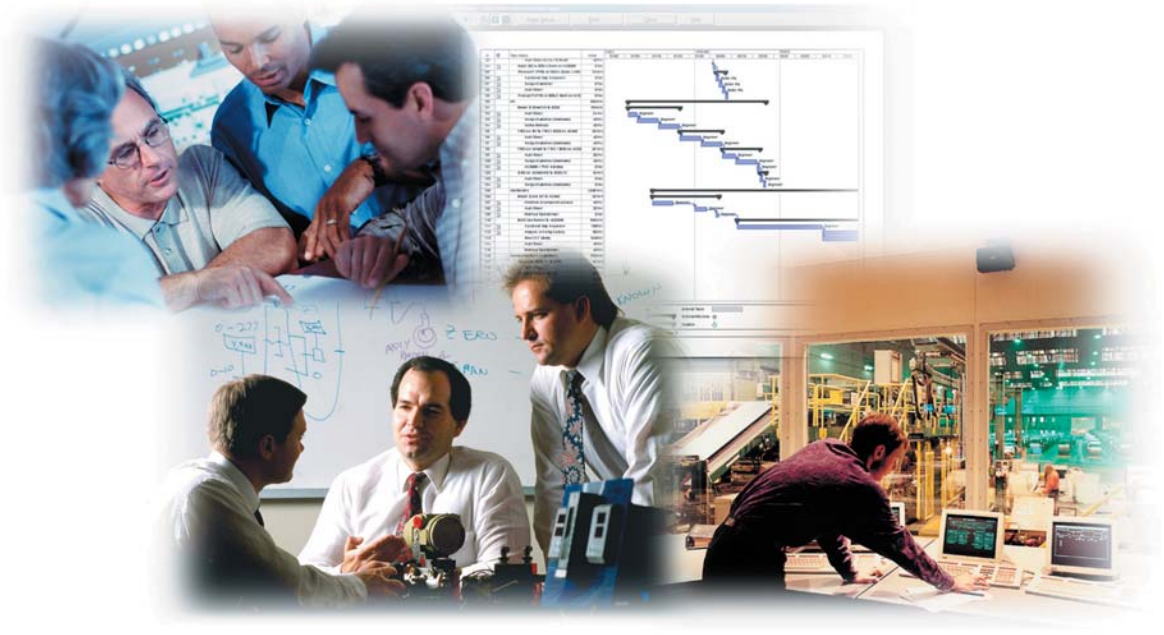

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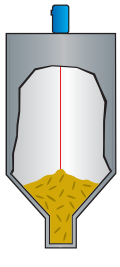
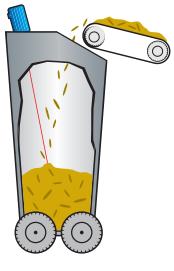
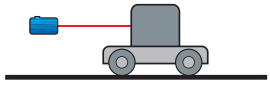

Measurement in Bins and Silos


Level Control in Roller Crushers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Global Warming: A Trillion-Dollar Hoax

Yes, the Earth is warming. It has been warming since 1850, the end of the Little Ice Age. Global warming is a natural phenomenon, something that happens periodically. The hoax being foisted upon us is that (a) it is our fault and (b) we can do something about it.

If the tree-huggers, proponents of the Kyoto Treaty and its subsequent Protocols, university researchers and Third-World countries get their way, the hoax will cost Americans and Europeans trillions of dollars. Developing nations, such as India and China, are almost totally exempt from the treaty, so refineries and chemical plants in North America and Europe will simply move more plants to exempt Third World countries. The loss of jobs and income will be staggering, and it won't stop global warming.

The research behind *State of Fear* completely debunks the junk science surrounding global warming.

We've been warning you about the junk science involved with global warming in this column for years. But anyone who opposes the idea that we can actually do something about global warming usually is summarily dismissed.

The research behind Michael Crichton's recent novel, *State of Fear,* completely debunks the junk science surrounding global warming. The book has more footnotes than I have ever seen in a novel, and it's backed up by 34 pages of author's notes, appendices and a bibliography listing resources, references and technical papers.

Crichton's message is:

- We know astonishingly little about every aspect of the environment.
- Nobody knows how much of the warming trend is natural and how much is man-made.
- Nobody knows how much warming will occur in the next century. Computer models vary by 400%. He predicts it will go up by 0.812436 °C, and says his guess is as good as anyone's.
- The oceans are not rising and all glaciers are not receding (some are, some aren't).
- He suspects that the real man-made contributor to global warming is land use (cutting down the rain forests, for example).
- Most important, he says, is that before we destroy our economies by passing dumb laws, we should develop accurate climate models.

Where have we heard that before? From our own control engineering expertise, which says you can't control something unless you understand how it will respond. Béla Lipták, my fellow CONTROL columnist, is writing a three-part analysis of global warming. Whatever he writes won't be junk science. If he says we can do something about global worming, I would believe it. Until then, I remain skeptical. To any control engineer, it makes no sense to pass restrictive environmental laws unless we know what the effects will be.

Don't get me wrong. I am in favor of cleaning up the air and the water, reducing emissions, eliminating smog, and stopping acid rain that destroys lakes. But let's do it for the right reasons, not because junk science says it will stop global warming.

How did we become so convinced that global warming can be stopped? Crichton says it's because global warming has become politicized. It is politically correct to be concerned about it, to accuse North America and Europe of causing it, and to propose laws to stop it. Woe to professors and scientists who disagree with global warming dogma. Their data is suppressed, their grant applications denied, and they don't get published. Crichton says that one proof of this is that so many outspoken critics of global warming are retired professors, who are no longer seeking grants.

Politicians usually go along with such trends, but not in this case. The U.S. Senate rejected 95-0 a non-binding resolution related to the Kyoto Treaty, which says it couldn't support an agreement that didn't tie developing and developed nations together in reducing greenhouse emissions. The U.S. House of Representatives also steadfastly refuses to pass laws that will cripple our industries. We are probably safe for the time being.

Nevertheless, it pays to be vigilant. Read Crichton's novel, and get better acquainted with the subject. It presents the case against global warming junk science, is an entertaining read, and gives you plenty to think about.

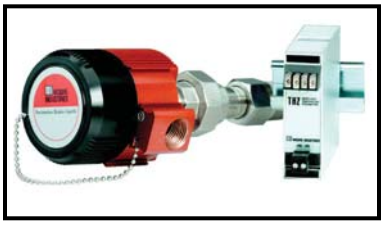
***Rich Merritt,* Senior Technical Editor**
rmerritt@putman.net

Blowing Hot and Cold: Focus on Temperature

CONTROL'S Monthly Resource Guide

TEMP TRANSMITTER BULLETIN

The technical bulletin, "Why Use Temperature Transmitters Instead of Direct Wiring?," details 10 cost-cutting, time-saving reasons for choosing a temperature transmitter in place of direct wiring to transfer data between field sensors and a PLC or DCS. Whether to protect signals from RFI/EMI, enhance accuracy and stability, or simplify engineering and maintenance, this bulletin covers all of the benefits. It can be downloaded at www.miinet.com/products/ca_temperature.shtml. **Moore Industries-International; 818/894-7111; www.miinet.com**



PICKING TEMP CONTROLS

The online "Controller Selection Tool" enables engineers to enter loop controller requirements in the order of importance based on their application. Radio buttons and drop-down menus allow selection of various criteria such as thermocouple input type, control output, alarms, approval bodies, sizes, etc. Users are then presented with specific controllers that meet their criteria, eliminating all the others. The tool is online at http://catalog.sensing.honeywell.com/imc/ss.asp?FAM=controllers. **Honeywell Process Solutions; 215/641-3798; www.honeywell.com**

FREE REFERENCE CD

The free "Databook VII" on CD has technical reference material such as thermocouple/RTD tables, temperature conversion charts, temperature controller selection considerations and practical guidelines for temperature measurement. It also includes a Transactions section with helpful articles. This interactive, multimedia CD can be viewed with an Internet web browser such as Microsoft Internet Explorer. The CD can be obtained at the web site below. **Newport Electronics; 714/540-4914 x358; www.newportUS.com**

MEAN KINETIC STREETS

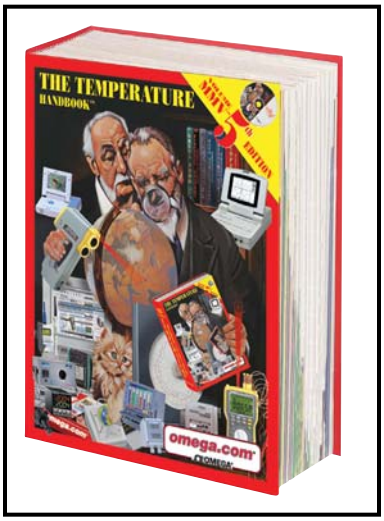
The "What is Mean Kinetic Temperature?" white paper describes a simplified way of expressing the overall effect of temperature fluctuations during storage or transit of perishable goods that must meet U.S. Food and Drug Administration (FDA) MKT standards. The two-page document explains the mathematics involved in calculating MKT. For a copy, send an e-mail to MTM@monarchinstrument.com or visit the website below. **Monarch Instrument USA; 603/883-3390; www.monarchinstrument.com**

TEMP/HUMIDITY MAPPING GUIDE

The free "Temperature and Temperature/Humidity Mapping Guide" provides step-by-step instructions on how to create and maintain effective facility mapping programs, including practical advice on how to determine critical mapping points, appropriate sampling rates, data logger selection criteria, and how to approach data analysis, remediation considerations and other topics. Download the guide at www.dicksonweb.com/article/article_26.php. **Dickson; 630/563-4222; www.dicksonweb.com**

TEMPERATURE WEB PAGE

These web-based temperature resources include: "Thermocouple Tutorial," "Thermocouple Wire Tutorial," "Temperature Measurement Guide," "Wire Color Codes and Limits of Error," "Infrared Thermometers," "Thermistors," "RTDs," "Temperature Controllers," "Thermocouple Assembly Quick Guide," "Temperature Measurement and Control," "Using Thermocouples," "Thermocouple Reference Data," "Thermocouple Tables," "Cryogenic," "Conversion Charts" and other useful information. Visit the web page at www.omega.com/techref/. **Omega Engineering; 203/359-7815; www.omega.com**



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Certainly, none of this year's inductees into the Process Automation Hall of Fame decided as children that they wanted to be Automation Heroes. Automation is a career that most people just fall into, as you will see in the careers of Dick Morley and Cy Rutledge. Nonetheless, these inductees' career accomplishments rank them in the small number of people, whose contributions to process automation have made significant differences in the art and practice of the discipline.

BY WALT BOYES, EDITOR IN CHIEF

We all need heroes, even automation engineers. Especially now, we need heroes in process automation that we can point children to

EXEMPLARY CAREERS PRODUCE

REAL AUTOMA

Edgar H. Bristol II

Ed Bristol is a Fellow Emeritus of the Foxboro Co. In fact, the fellowship he holds is eponymous—the Bristol Fellows are named after him. Since November 1959, he's been associated with Foxboro, and is responsible for many of the control innovations that have emanated from that vendor for nearly 50 years.

Currently, Bristol is striving to avoid retirement, and working as a consultant, both inside Foxboro and outside. With fellow Hall of Famer Ted Williams, of Purdue University, Ed had a leading role in creating the "Purdue Workshop for Industrial Computer Systems" from which many continuous and batch process control techniques have come. He is a well-known author, who has well over 100 papers, and a dozen patents to his name in control, adaptive control, multivariable control, control software and languages.

Like fellow inductee, Dick Morley, Bristol is a graduate of MIT. He is the originator of relative gain array (RGA) analysis (1965), which is the forerunner of today's multivariable control. He also originated pattern-recognition-based adaptive control strategies that are used widely today, especially in machine vision applications.

Bristol is a tireless contributor to professional societies like ISA, IEEE, AIChE, ACM and others, and has been active in standards writing efforts for nearly 40 years. Among his other awards, appropriately, is the Dr. Guido Carlo-Stella Award, which he received from WBF in 2004, for his seminal effort in getting WBF started and his contribution to batch theory.

One of Bristol's most important recent contributions to the field of process automation is the paper he delivered at ISA 2002 entitled, "A Control Retrospective: What We Should Have Done." The conclusion of the paper says, "Process Control is in a quandary. The digital systems, which should make life easier, continue to make it harder. And at a time when digital expertise is beginning to face up to the consequences of digital complexity [sic]."

Bristol goes on to say, "A restoration of the traditional user/vendor division of labor would act to restore a real progress to the field." Whether he is right or not, Bristol's exemplary contribution to the field of process automation goes on.

Ed and his wife, Julie, have three daughters. He also enjoys playing with rhododendrons, orchids and bicycles, when he isn't doing process automation.

when we talk about the benefits of careers in manufacturing and automation. Fewer and fewer children are interested in manufacturing as a career choice. Not all manufacturing is going to be off-shored, and we'll need those young people in the years to come.

CONTROL is proud to present to you this year's class of inductees into the Process Automation Hall of Fame. The induction will take place, once again, during the Hall of Fame dinner at the World Batch Forum (WBF) on March 5 at the Sheraton Gateway Hotel, near Atlanta's Hartsfield-Jackson International Airport. You can get all the details at www.wbf.org. We invite all of CONTROL's readers to attend, and to stay and attend WBF.

PROCESS AUTOMATION FAME

ATION HEROES!

Richard E. Morley

CONTROL's readers know Dick Morley as a frequent contributor to the "Other Voices" column in this magazine, but he is far more than that. It is a rare bird indeed who contributes to more than one life-changing invention, but among Morely's patents are four that are indeed life- and culture-changing.

In 1960, he patented magnetic thin film. His invention made possible high capacity data storage, first on floppy disks, then on tiny hard drives, and even on those found today in iPods. In 1968, he founded Modicon, and (along with the late Otto Struger, of Allen-Bradley) is considered the "Father of the PLC." In 1970, he patented a handheld terminal (Termiflex) that is the ancestor of field calibrators, programmers and the PDA. Finally, in 1976, he founded Andover Controls, and created the first modern building automation system.

While all these patents are used in many industries, they all have had a substantial impact on process automation. As the PLC matured from a "relay replacement" to a complete, hybrid controller, the modern distributed control system in process automation began to look a lot like a PLC with an HMI at the other end of a network. Nearly every instrument vendor provides a handheld programmer that is a direct descendant of the Termiflex terminal.

Since then, Morely has had a hand in founding more than a dozen companies, and serves as a very popular high technology consultant to business and government. Recently, he made a new career as a manufacturing futurist, and serves as board chairman for the National Center for Manufacturing Sciences (NCMS).

Morely also is a subscriber to the "give back and pay forward" school of life. He and his wife Shirley have had over 30 foster children during their 46 years of marriage, and he is a major driver in an angel investment group that targets young entrepreneurs for help.

While it is not true that Morely quit his real job to become a consultant because his employer insisted that he wear a suit and tie to work, it is true that he quit because his employer wouldn't allow him flextime, so he could ski during the week and avoid the lift lines. Morely is a Harley-Davidson rider from way back, riding almost immediately after knee surgery, and he also has a marvelously customized John Deere tractor, which he can drive over the Web.

Wyman "Cy" Rutledge

"In the early days of process control," Wyman "Cy" Rutledge remembers, "we began applying this method to papermaking at Mead Paper Corp. (now MeadWestvaco)." Rutledge was principal scientist for 35 years at Mead Research and Development in Chillicothe, Ohio. During that time, his contributions to controlling the process of papermaking were enormous. Rutledge also taught math and physics at several universities in Ohio and New York, while continuing his career at Mead.

"We were familiar with the control work done by IBM and a steel company in Cleveland, and thought this could be applied to paper processes," says Rutledge. "It was used for applying color, coatings, fillers and other additives in the papermaking process. It also was extended to controlling the basis weight by means of a variable stock flow valve at the headbox."

The information was fed back to controllers and the lab from a basis weight measurement at the reel. This information was used for studies, not just for feedback or feedforward control. According to Rutledge, this enabled Mead to move ahead of its competitors by supplying higher-quality paper to companies like Hallmark.

Rutledge then participated in installing one of the first computer-controlled paper machines, and the first one at Mead.

Over the years, until his retirement in 1990, Rutledge became one of the most effective theorists of process control as applied to papermaking, and his contributions to professional standards bodies are considerable. He contributed both to continuous process management and to batch process management, especially in the addition and creation of additives. He was one of the principal founders and supporters of Miami University of Ohio's Pulp and Paper Symposium, and received an honorary doctorate from the university.

Rutledge graduated from Hiram College in Hiram, Ohio, served in the U.S. Army during World War II, and earned both a M.S. and Ph.D. in nuclear physics from the University of Michigan at Ann Arbor. He is, therefore, another lifetime process automation professional, who initially set out to do something entirely different—a typical occurrence in the field.

Rutledge continues to live in Chillicothe with his wife Mary, but his increasingly poor health in recent years will prevent him from coming to WBF to receive his award in person.

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Kathleen Waters

Process automation is now old enough to have more than one generation of heroes. Ed Bristol, Dick Morley and Cy Rutledge are from the first generation of modern process automation, while Kathleen Waters is a second-generation process automation hero. She has been a principal engineer with Genentech since 1991, and was previously with G. D. Searle (which became NutraSweet in 1984) from 1982 until 1991.

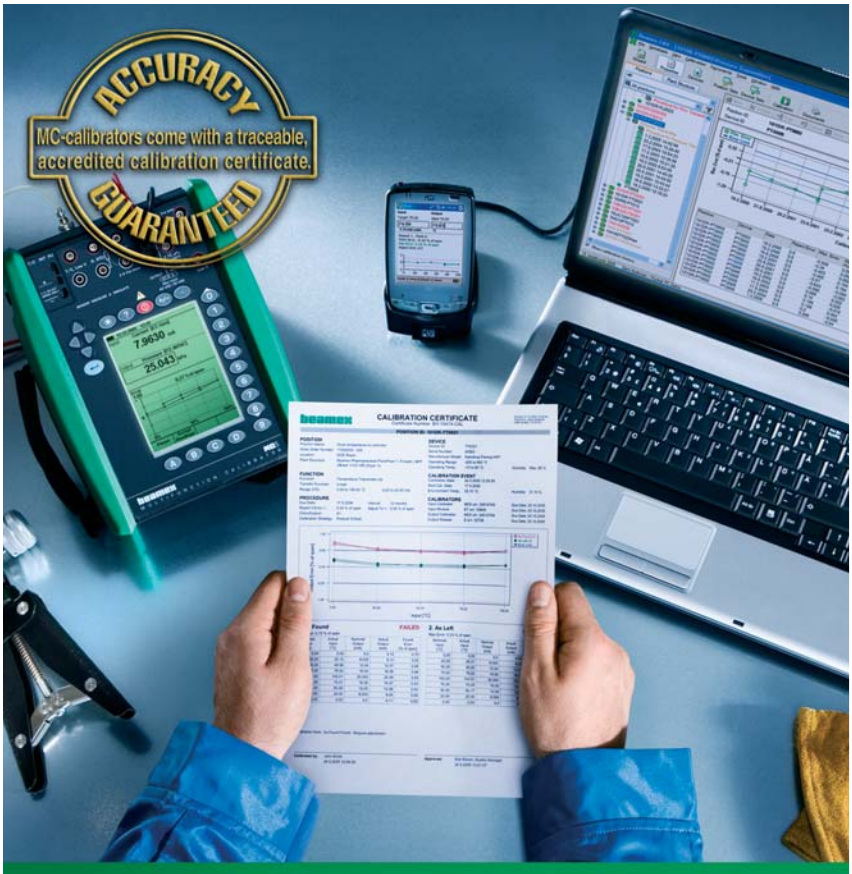
Waters has been responsible for many strategically important assignments in her career, including process systems software development, process information management systems, manufacturing execution systems, automation architecture and quality initiatives for systems supporting process development and cGMP operations. Note that her career *starts* where the groundbreaking developments of the first generation of automation heroes leave off.

Waters established a scalable automation architecture, enabling process and implementation standardization. She was an early adopter of plant-floor Ethernet, automation network segregation, and web-based data access. Her work relies heavily on industrial standards and benchmarking to establish models for implementation. She has worked repeatedly on compliance programs, collaboration and driving best practices.

Waters also led development of the S88.04 Batch Production Record standard, a reference model for batch production records, containing information about production of batches or other elements of production, and she currently is the ISA S88 committee's vice chair, as well as a member of WBF.

In addition, Waters' experience with batch processes gave her the ability to make S88.04 a true cross-industry standard, which has been adopted, not only by batch processing companies, but also by automation vendors, MES and ERP departments, and even by vendors like Microsoft, IBM and SAP. Her infectious energy, enthusiasm and sense of humor have made it possible for her to lead teams of disparate constituents to common goals and solutions. She listens well, which allows her to tread the minefields of modern standards efforts, and produce timely and relevant solutions to major process automation problems.

Waters loves to golf, ski, and like other Hall of Fame members, to garden. Oh, yes, one final thing about Waters. We're pleased to note that she is the first woman to be inducted to the Process Automation Hall of Fame.



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IS FDT GOOD FOR FIELDBUS AND DEVICE NETWORKS?

Field Device Tool/Device Type Manager is at the Center of a Controversy—Is This Fieldbus Wars II? Or Can FDT/DTM and EDDL Work Together? **by Ian Verhappen, Contributing Editor**

Field Device Tool/Device Type Manager (FDT/DTM) is a multi-network-compatible, device vendor-defined set of software tools that allow end users to view and interact with a device using the vendor's own software. Since the technology was introduced at Interkama in 2001, improvements have been made to it. Of course, the underlying electronic device description language (EDDL) technology on which it relies has been improved as well.

These two technologies also have been at the heart of a controversy involving instrument suppliers and users. The main question is: do these two technologies conflict, or can they work together to complement each other? It appears most vendors and users have finally agreed that they do indeed complement each other.

To understand how these two technologies are interrelated, let's start with a short lesson on how each of them works.

EDDL was created at the dawn of fieldbus for the purpose its name implies—it is used to describe devices. Systems that interact with fieldbus devices need to know the rules of the conversation before communication can start. What function blocks are present in a device type? What parameters are available? What are the data types of those parameters? What are the default values and permitted ranges of those parameters? This information is used by the system to understand a device even before the device is present in the system. As a result, fieldbus technologies, from the communications specifications through the various device descrip-

What Is FDT/DTM Anyway?

FDT/DTM technology is made up of two basic components:

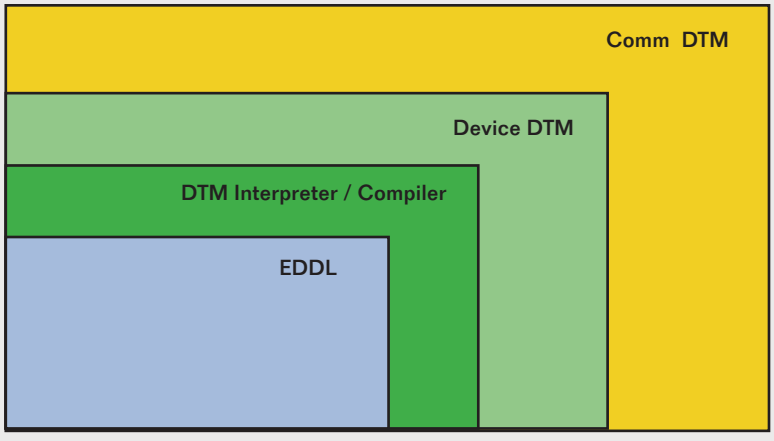
- The Frame Application, which is the Device Management Environment.
- The Device Type Manager (DTM), which is a software component that simply operates as a "driver."

DTMs can be further classified into two categories:

- CommDTMs are software components/applications that enable the Device Management Environment, called a Frame Application, to access different fieldbuses. This approach makes the Device Management Environment open to all fieldbuses. Just add the required CommDTM. CommDTMs for HART, Profibus, Foundation Fieldbus' transducer blocks and Interbus S already exist, and are being developed for AS-interface, DeviceNet, Profinet and Modbus.
- Device DTMs again are software components/applications, but these DTMs enable users to configure the associated device within the Device Management Environment. These DTMs are the individual "device drivers" that allow users to configure, diagnose, and manage the field device. The DTM is often compared to the drivers loaded into computer operating systems to support external devices such as printers. These don't just to allow the printer to operate, but also help diagnose and troubleshoot faults (ink cartridge low, align print head, etc.). The DTM offers this same support for field devices connected to a control system (control valve stuck, signature change, etc.).

Figure 1 shows how EDDL and FDT/DTMs combine to communicate between the field device and the "host-based" software, be it a configuration tool or asset management system.

THE MATING DANCE OF EDDL AND FDT/DTM



Combining to communicate, whether it's a configuration tool or an asset management system.

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tions, look at communications from the device's perspective.

FDT is device, protocol and system independent, and therefore is capable of managing all digital communications protocols. FDT is the interface specification, while DTM is a driver that presents an interface to the physical communication channel between the system or tool and the actual fieldbus devices. The second component, the DTM frame application, is an interface to the platform.

FIGURE 1.

THE MARK OF THE BEAST?

Some DCS vendors froth at the mouth when they see this symbol, but end users welcome it as a symbol of open standards.

End-User Perspective

When asked about the uses of FDT/DTM, many end users reply, "Ultimately, what it boils down to is what can be realized in the end-user world?" One end-user, who asked not to be identified because he felt that it might compromise his company's relationship with its vendor, adds bluntly, "My hope is that vendors will use the right tool for the job rather than trying to win a political battle over market segmentation."

An engineer at Shell Global Solutions responded that they would like to design a test that demonstrates EDDL's value by itself, and compares it to a combined EDDL and FDT/DTM implementation. His thought is to take one or two typical positioners with their valve diagnostic features and one or two systems, and demonstrate how the diagnostics and advanced diagnostic information from the devices can be integrated into the DCS and asset management systems, either via EDDL or via the combined EDDL-FDT/DTM route. "This will give the end-user community the insight as to what approach provides the best value for its money," he says.

In fact, some people believe that this is the heart of the controversy. Jim Sprague, who works for a major oil and gas producer in the Middle East, concurs with this idea. "We want to be able to use best-in-class Foundation Fieldbus instruments, regardless of vendor, with the host DCS of our choice," says Sprague. "We also want to be able to access all maintenance, configuration and diagnostic software and displays—for 'complex' instruments—on our DCS system maintenance stations.

"We expect this software and its displays will operate and look/feel as the OEM instrument vendor intended. This software needs to be well integrated with the DCS, so it communicates configuration changes to the host system database, as well as manages online and out of service (OSS) mode changes appropriately."

End-User as Agnostic

Sprague adds, "We don't care if this software works via EDDL or FDT/DTM. We expect our host-system vendors will provide this interface, and will help us keep it current. We don't expect a problem with either technology, and we don't see that they will compete. We expect our selected host-DCS vendor will support both these technologies, and help us apply whichever works appropriately for specific applications.

"At present, the only place we use this kind of complex field device interface is on our control valve positioners. We use 'plug-ins' and 'snap-ons' from the positioner vendor that allow access to their maintenance and configuration software from our selected host DCS system."

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"Aye, There's the Rub!"

"These applications are limited by which vendors have developed these plug-ins and snap-ons for which host-DCS systems, and the choices are few," says Sprague bluntly. "We look forward to the day when all complex instruments have their software integrated by either EDDL or FDT/DTM. Complex instruments we're looking to integrate in this way including valve positioners, contact/non-contact radar level instruments, simple and complex analyzers, MOV valves, multi-variable transmitters and others."

Another anonymous end user from an international petroleum company that is relying heavily on Foundation Fieldbus for all its control system growth adds, "I don't think FDT/DTM is a good solution for device configuration. This is where I think EDDL is best. I also think FDT/DTM technology can be of use in advanced diagnostics. I think there are some challenges to work through in this regard, but the potential is there."

Challenges for FDT/DTM

FDT is an executable program that is provided by the device vendor, and is downloaded off the Internet to be loaded or installed on your computer system. Because DTMs are multi-megabyte files, it's not possible to load them into field devices. Also, because these files are executables downloaded from the Internet, some users will have concerns about their integrity, and consider them to be possible security risks.

"These applications are limited by which vendors have developed these plug-ins and snap-ons for which host-DCS systems, and the choices are few."

Plug-Ins and Snap-Ons

Interestingly enough, almost all of the above mentioned plug-ins and snap-ons rely on COM/DCOM just like FDT/DTM does, but are only useable with the host/asset management system for which they're written. Consequently, they have less functionality than the open standard-based FDT/DTM systems. As of October 2005, the DTMs are subject to certification by the FDT Joint Interest Group (JIG). A list of certified DTMs can be viewed on the FDT-JIG web site at www.fdtgroup.org. All certified FDTs have an "approval check mark" (Figure 1), which is similar to the one used by the Fieldbus Foundation (www.fieldbus.org).

In addition, FDT/DTM reportedly provides clearer responsibility in projects. The responsibility for the interfacing to the device is not floating between device vendor and DCS-host vendor. A lead engineer working at a major European company reports his plug-ins (a dedicated valve package) didn't work after the host-vendor's asset management package was upgraded to the next level. It's then difficult to get the device vendor to come in and modify the interfacing just to get the same functionality again.

There also has been discussion of FDT on the Fieldbus Forum, where the concern is that using ActiveX could pose additional security risks to a system, or at least give someone an opportunity to use the technology to launch a cyber-attack on associated systems. This is because FDT/DTM requires the necessary ports and open access to work.

Of course, if an outsider can access the network on which this information is being transmitted, the system likely has bigger problems than ActiveX to worry about, and needs a complete security audit. This system should not be open to the business LAN or public internet environment in the first place.

The FDT engine runs on top of OPC DA, and uses DCOM to get data from controllers. OPC DA is not optimized for this high level of traffic and communications. However, as OPC Univer-

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sal Architecture (UA) becomes available and this technology becomes adopted, reliance on DCOM will be removed. Interestingly, EDDL is also working with the OPC Foundaton (www.opcfoundation.org) to move to a UA environment.

However, running diagnostics in the host environment is limited by bandwidth of the communication channel to field devices. This limitation will restrict the diagnostic performance possible in the host versus running the diagnostic in the field device.

Because the host system reads the EDDL file directly, device configuration and control related interfacing will be much better integrated with the system when done with EDDL. At a recent meeting in Cambridge, Mass., Scott Bump of Invensys, and a member of the FDT Group, stated, "EDDL is for control and FDT/DTM is for applications," confirming FDT Group's position that these two technologies are complementary. As indicated above by the end users, there is hope that the organizations supporting these two technologies realize it as well, and will work together for better integration and application of the right tool for the right task.

Positives for FDT/DTM?

FDT reportedly will provide bigger benefits to the "slower" digital communications technologies, such as HART, by giving them a degree of advanced diagnostics capabilities, especially for off-line tools such as those associated with valves. HART Communications Foundation (www.hartcomm.org) is more accepting of FDT's capabilities because the devices are restricted to 9.6 kbaud, use ASIC processors for whichever functionality can't be added without a new chip set, and, after using the majority of their power budget to transmit the signal, have little left to perform the types of diagnostics being expected of digital devices.

A lead engineer at Shell Global Solutions suggested that a big positive of FDT is, "They managed to get the HART and Fieldbus Foundations moving on DD improvements—albeit rather late in the game." These enhanced EDDL improvements were demonstrated at the recent ISA Expo 2005 show by the Fieldbus Foundation, while more work is in development.

End-Users Wanted

Taking a page from the success of end user councils, originally established by the Fieldbus Foundation, and now adopted by other groups including OPC, FDT Group held its first End User Group Forum in October 2005 in Amsterdam. The event was attended by more than 40 participants, including representatives from 14 end user and engineering companies. Those attending the event learned more about FDT technology, and also provided input on requirements to make it better in the future. Based on the success of this initial meeting, similar events will be held in 2006.

FDT also offers solutions for "Nested Networks." It offers the chance to diagnose and monitor a remote I/O card's health centrally, using one fieldbus technology. At the same time, FDT drills down to a device on the other side of the remote I/O card, where the device is typically operating on a different fieldbus or protocol. Simplifying Nested Networks enables a centralized maintenance strategy to manage, configure, and diagnose field devices on a system with multiple different protocols. This is not an uncommon situation in today's modern digital control systems.

Because FDT/DTM is an open standard, it enables non-DCS device vendors with a level playing field for all device manufacturers.

Gareth Johnston, fieldbus communications specialist at ABB Ltd. (www.abb.com), states, "The value of FDT is that FDT provides full access to intelligent device features such as set-up (especially for multi-variable devices) and diagnostic information (signature analysis—clear language maintenance description). The DTM can use these to execute automatic commands to simplify device set-up. It also provides a flexible communication path that allows users to select the communication device (remote I/O with HART Passthrough, USB HART/ProfiBus modem, etc.) regardless of the vendor."

For devices that don't have a DTM yet, there also are FDT applications on the market that interpret EDDL applications on the fly. This makes it possible for end users buying equipment without FDT support to generate a basic FDT for use in their asset management or host system.

Because FDT/DTM is an open standard, it enables non-DCS device vendors with a level playing field for all device manufacturers. This helps them compete more strongly, using their enhanced diagnostics capabilities, and providing full access to these capabilities in any environment. Unless the snap-ons and related "add-in" pieces of software used to communicate with the asset management system are made available as an open standard, as was done with DD technology and is being proffered by FDT Group, the owner of the software for which the add-in is developed will remain in control of the way data is presented and made available.

The recently released FDT Style Guide will provide a more consistent "look and feel" for future DTMs, while also satisfying the recommendations of the NAMUR working group on the features/requirements of an effective asset management system. As a result, it's likely that most DTMs will provide a similar "look-and-feel" device interface that is very similar to the vendors' existing maintenance and configuration software tools.

What's in the Future?

Most control system and device vendors are looking at FDT/DTM to see if or how it fits into their future. It looks like it may be an even better fit after OPC UA is available, though most of the vendors will likely continue to use EDDL for configuration.

Yet another anonymous major oil company engineer is concerned that, "Users will have to maintain a steady push to assure that diagnostic data and alarms are integrated with DCS operator subsystems for effective graphical user interface and alarm management. Vendors have a distressing tendency to segregate these applications into separate systems that don't share information with operators."

In addition, after the Swiss National Standardization recently made a new work proposal to the International Electrotechnical Commission (IEC, www.iec.ch), FDT was unanimously approved as a "publicly available specification" under the auspices of new standard working group IEC SC65C WG14 "FDT." Following an initial meeting in January 2006, the hope of the committee is that the standard will be completed by October 2007.

Similarly, Profibus Nutzerorganisation e.V. (PNO) and FDT Group have agreed to develop a PLC programming tool, Tool Calling Interface (TCI), as an open tool to integrate the factory automation and process automation environments with one interface. The TCI specification will contain a bidirectional communication channel from the tool to the field device via the PLC programming system. Hopefully, this tool will also be compatible with the IEC 61508 on which all PLC programming software and documentation is based.

Also, FDT-JIG has indicated that FDT is a complementary rather than competitive technology to EDDL. They also acknowledge that FDT has some limitations including:

- FDT is a Microsoft Windows-based technology. As such, it is subject to the inevitable upgrades and operating system version changes with which we have all become familiar. As with all software tools on all operating systems, FDT platforms and device applications will need to be updated occasionally.
- FDT only defines interfaces between system components. As such, FDT components cannot replace DD files, which continue to be an integral part of major fieldbus systems.There is hope that the FDT/DTM versus EDDL battlesoon may be declared a ceasefire, since a member of the Fieldbus Foundation's (FF) board of directors says, "Most of the members of foundation are also members of FDT Group, so the board needs to make a clear statement on the EDDL/FDT relationship." Hopefully this can happen at the FF's general assembly at the end of February 2005.

Using FDT

Achim Laubenstein, ABB's manager of fieldbus development, says, "FDT is the opportunity for device vendors to provide the application software that best fits their device. One and the same DTM can be used in many hosts without extra effort. This saves costs for the host and on the device side. One can imagine that device vendors especially have a high level of interest in such interface technology.

"When implementing FDT, rather than starting from scratch, we recommend reusing existing components available on the market. A user should check whether the products he wants to use are certified by FDT Group."

End-users just want a simple, intuitive way to access, interpret, and understand what's happening in their process and to their devices. Fortunately, it sounds like the manufacturers also are getting the same message, even though it looks like some are fighting a rear-guard action to protect proprietary technologies from the open systems movement. Hopefully, OPC UA will be the common technology that will enable them to come together in a smooth, unified way.

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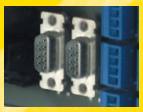

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ZAP! HOW DO IT KNOW?

Laser-Level Technology Expands Options in Applications and Environments

by David W. Spitzer, Spitzer & Boyes LLC

Non-contact, level-measurement sensors were developed for applications where contacting sensors were seen as problems, either through false readings or high maintenance.

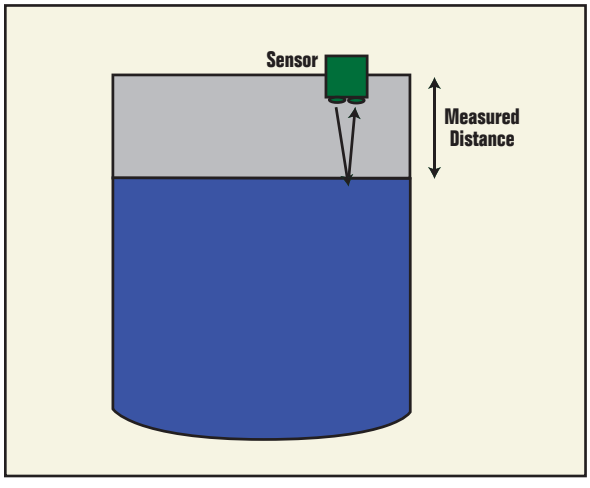
All non-contact, level-measurement systems consist of a level sensor located above the material surface that emits a signal, and processes the returns (reflections) of that signal from one point in the vessel. This is not a problem where the gas/liquid interface is horizontal, but multiple interfaces affect the performance of ultrasonic-level devices, while low-dielectric-constant interfaces affect the performance of radar-level devices.

In solids applications, material entering and leaving the vessel affects the solid/gas interface. For example, a rat-hole may form as solid material leaves the vessel. If the level measurement reflects a point in the rat-hole, the measured level will decrease (as expected). However, if the material remaining on the sides of the vessel falls and fills in the rat-hole, the level will abruptly (and unexpectedly) increase.

Laser-level sensors emit a laser beam toward the material, and measure the level of the material by measuring the time that the laser beam takes to travel to and return from the material. The distance between the sensor and the material is one-half of the measured time-of-flight multiplied by the speed of the laser beam. Mechanical dimensions can then be used to determine the percent level in the vessel.

FIGURE 1.

LASER LEVEL APPLICATIONS



Lasers can be used in many non-contact applications where radar and ultrasonics don't work.

Laser-level technology isn't limited by the dielectric constant of the material (like radar) nor its propagation velocity in the vapor space (like ultrasonic), and the laser beam doesn't diverge much, so it can target smaller areas than radar and ultrasonics. Further, in some applications, lasers can be aimed to sense the level in locations that can be difficult to measure using other technologies, such as inside the chute of a bin.

Degradation can occur at the sensor, in transit, and at the surface of the material, resulting in failure. Dirt, dust or other coatings on transceivers can make the received laser signal weak. Foam conditions may vary over time, and cause erratic level measurements.

Many laser sensors use safer Class 1 lasers. Others use Class 3 lasers that can pose a risk of eye injury

Jim Van Rems, vice president of Riegl Laser Measurement Systems (www.riegl.com), says, "Laser-level systems are among the most expensive technologies available, so they tend to serve niche markets where no other is viable." Measuring the level of molten glass and metals at temperatures as high as 2,000 °C is one such application, according to Van Rems.

Other typical applications (Figure 1) include measuring the level of material in tall tanks that pose problems for radar and ultrasonic level technologies. Van Rems also warns against installations where the vapor space is opaque to light and/or contains fine particles, reflective surfaces and/or electric charges. For example, the level of flour is difficult to measure because it contains fine particles that reflect light and contain an electric charge that causes light to be obstructed, reflected, and collect on the sensor. Steam is another example of a vapor that reflects light.

Pneumatically filled vessels pose more problems than mechanically filled vessels because they tend to disperse particles in the vapor space, which can hinder a laser-level measurement system. Van Rems cites a rule of thumb: "If you can see the level despite these influences, laser-level technology will work."

Ray Charron, product sales specialist with Optech (www.optech.ca) adds, "Vapors with coarser particles are usually more easily penetrated than vapors with fine particles.

Darker particles are easier to penetrate than lighter particles that tend to reflect the laser energy."

Compensation for color and reflectivity are often necessary to accurately measure level, especially for high-temperature applications like molten steel, where the color and appearance of the metal can change in a matter of minutes.

Laser energy typically doesn't penetrate foam to measure the level of the liquid. Charron adds that laser measurements generally benefit from diffusion on the material surface, and that surfaces that reflect too much laser energy can saturate the receiver.

"If you can see the level...the laser-level technology will work."

Ivo Radanov, laser-level product manager at K-Tek (www.ktekcorp.com) points out, "Laser-level measurements are independent of the angle of the material because part of the scattered reflection will return to the sensor."

In some applications, this scattering will completely defeat the measurement method, but Radanov adds, "This provides an advantage over other non-contact technologies that use energy that can be reflected away from the sensor measuring the reflected energy."

Evert Pol, preventive predictive maintenance and reliability engineer, Dow Chemical Co. (Temeuzen, Netherlands) uses several laser systems for granular materials. "These instruments work well, after start-up problems, and are versatile in that they can measure from the top to the bottom of the silo." Pol warns other end-users, "Be careful to consider a dirt protection tube that addresses the presence of dust."

Pat White, director of training at Weyerhaeuser (Simsboro, La.), gives laser level high marks, but not without some of the pain associated with applying new technology. "When first installed, fines in the vapor space near the transmitter caused false levels at times," White says. White solved that problem by installing a hood over the level transmitter to keep the fines from getting on the transmitter. In the end, the manufacturer developed software that could ignore the fines.

Laser-level measurements have become viable recently enough that there are relatively few suppliers. Time will tell if it will give end-users another tool in the battle for productivity.

David W. Spitzer is a principal in Spitzer and Boyes LLC, which offers consulting services for the process industries. He can be reached at 845/623-1830 or at spitzer@spitzerandboyes.com.

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TO SAFETY STANDARDS AND BEYOND!

In 2004, ISA adopted the most recent safety standard, ANSI/ISA 84.00.01, which we call S84.

by William L. Mostia, Jr., PE, principal, WLM Engineering Co.

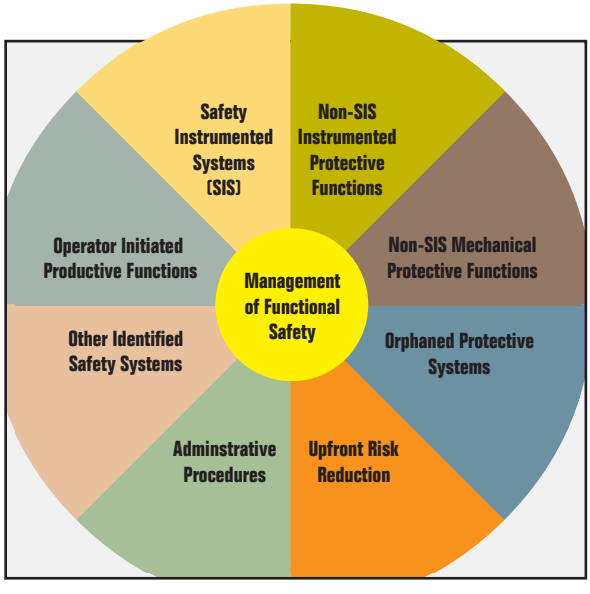
The current implementation status of S84 is mixed. Some companies have embraced the new standard, while others have adopted varying degrees of it.

It appears for new projects, particularly those done by E&C contractors and outside consultants, that most companies are implementing at least parts of S84. The main parts being addressed appear to be safety integrity level (SIL) determination, SIL verification, safety requirements specification (SRS) development, and test plans. The record is spottier for other aspects, such as the safety instrumented system (SIS) safety-lifecycle management requirements, implementation of the safety lifecycle, equipment selection requirements, competence requirements, training and procedural requirements, downstream verification and compliance auditing.

It seems many companies are managing SIS on a per-project basis. And some may not grasp the overall management and requirements required to consistently, successfully implement and sustain SISs and related systems.

FIGURE 1.

MANAGEMENT OF FUNCTIONAL SAFETY



An overview of functional safety management shows it includes safety, protective, risk reduction, and administrative systems.

For existing installations, progress appears somewhat less. While there is some increase in addressing existing systems, it appears to lag application of S84 to new projects. Some companies appear to be relying on the grandfather clause to help existing installations comply (see "The Grandfather Clause Is Not a Jolly Fat Man in a Red Suit," by Angela E. Summers, PhD, PE, CONTROL, Aug. '05, p. 68). Other firms are waiting until their systems are upgraded (with or without a plan), while some seem to be ignoring the issue out of ignorance or due to managerial decisions.

Grandfathering based on prior existence of a safety system does not satisfy the grandfather clauses. Grandfathering requires evaluation and documentation.

In the case of grandfathering, the problem is that some of these companies aren't truly complying with the grandfather clause, either in S84 or PSM, which essentially have the same requirements, but differ somewhat in terminology and extent. For safety systems designed and constructed in accordance with codes, standards or practices before the standard or regulation was issued, both grandfather clauses require the owner/operator to determine that the equipment is designed, maintained, inspected, tested, and operating in a safe manner.

It should be noted that this is not a passive statement. For example, grandfathering based solely on prior existence and operation of a safety system does not satisfy the grandfather clauses. Grandfathering requires that the safety systems be evaluated and documented, and that all requirements of the grandfathering (design, maintenance, inspection, testing, and safe operation) have been met. In addition, S84's grandfather clause applies to only SISs, while PSM's grandfather clause applies to SISs, identified non-SIS instrumented safety systems, mechanical safety layers of protection, and other process safety-related systems. If you don't satisfy the grandfather clauses, then the safety systems have to meet PSM and current relevant standards.

Layer of Protection Analysis

The current S84 standard provides guidelines regarding risk assessment, including details of various methodologies contained in its Part 3. The most common form of risk assessment in the U.S. is some variation of layer of protection analysis (LOPA). In Europe, the risk graph has a larger following, though some people do use the LOPA technique. We're going to assume that some form of LOPA has been used. If not, the concepts should apply to other risk assessment techniques. We're using LOPA as a generic term to refer to its various varieties.

In the LOPA method, you evaluate the unmitigated risk (frequency and consequence), and determine the risk reduction required to decrease the risk to a tolerable level. This is commonly done with a risk matrix, which plots consequence against frequency to give the desired risk reduction. The required risk reduction is then balanced against the existing (or designed) identified independent layers of protection (IPL). The properties of an IPL are well defined. Traditionally they are independence, specificity, reliability and auditability, but management of change and security should be added as well.

Once this has been done, any residual risk must be handled by adding additional IPLs to further reduce the risk to the tolerable level.

When the IPL involved is a safety instrumented system (SIS), S84 has requirements designed to meet governmental regulations and industry practice. When the IPL involved is not a SIS, this issue is somewhat murkier.

It's clear that safety IPLs are covered by PSM and environmental IPLs by RMP regulations, which provide requirements, but how exactly this is to be done is well not detailed. On one hand, SIS requirements are well detailed, but other IPLs, which are byproducts of the SIS safety lifecycle, are not well defined. LOPA's connection to non-SIS IPLs to the mechanical integrity or other management programs isn't covered by any standard. For example, the assurance that changes to a non-SIS IPL that affect its integrity, availability, or any properties of an IPL is based on an evaluation of the effect on the IPLs identified for a particular hazard and in light of the originating LOPA.

This clearly indicates that a site must have a system to manage IPLs to assure their integrity, availability and their inherent properties. A common method for doing this in instrumented systems is the critical alarm list or the critical instrument list. However, "critical" often isn't

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In 1996, a major milestone in process safety was achieved when the requirements for safety instrumented systems (SISs), such as emergency shutdown systems, ESD, safety interlocks and safety systems, were codified in ANSI/ISA S84.01-1996, "Application of Safety Instrumented Systems for the Process Industry."

In March 2000, OSHA officially recognized S84 as "recognized and generally accepted good engineering practice" for meeting the Process Safety Management (PSM) regulation CFR 1910.119 for safety instrumented systems.

During 1999-2002, the International Electrotechnical Commission (IEC) issued its standard 61508, "Functional safety of electrical/electronic/programmable electronic safety-related systems," which provided an umbrella standard for safety systems for all types of industries.

In 2003-04, IEC 61511, "Functional Safety: Safety Instrumented Systems for the Process Industry Sector," was adopted as the international standard for safety instrumented systems in the process industries.

In 2004, ISA and subsequently ANSI adopted the IEC 61511 standard with the addition of a grandfather clause and some conversion of text to ANSI/ISA 84.00.01 from 61511 as the new version of ANSI/ISA 84.00.01-2004.

For brevity's sake, the term S84 will be used in further text, unless it's more appropriate to use the full name. The new S84 standard brought with it a much more comprehensive standard that covered many of the PSM requirements, as well as substantially more management and documentation requirements. It also mentions covering environmental requirements in Section 1 (j).

well defined, and you get a large list that includes safety, environmental, asset protection, operational or other "critical" defined criteria. There is an ISA technical report, TR-91.00.02, "Criticality Classification Guideline," which provides some guidance in classifying loops. The difficulty is that there's typically no direct connection between LOPA and the critical instrument list. Just because it's on the critical instrument list doesn't make it an IPL and vice versa. Also, it's seldom identified in the list as an IPL, and may only be listed as critical and possibly describe what its criticality is. Simply being on a critical instrument list is not sufficient, without procedures and practices, to insure that all the requirements of an IPL are met and maintained throughout its life.

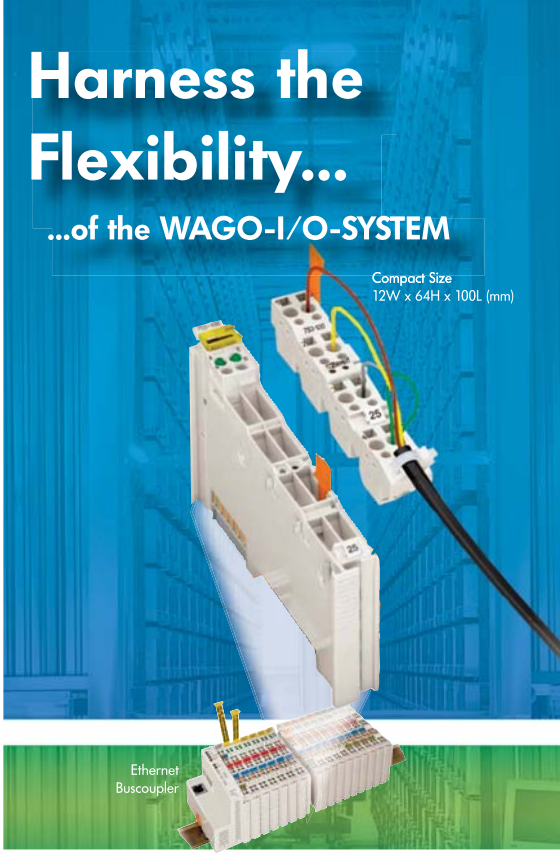
Hopefully, the way non-SIS mechanical IPLs are commonly handled is by placing them in the site's mechanical integrity program. This is better than nothing, though integrity programs aren't generally set up to handle IPLs and their requirements. Administrative IPLs usually are left up to plant personnel, and again have limited connection back to originating IPL requirements and the associated LOPA. More than likely, a lot of IPLs get lost in the shuffle.

Minimum Standards

It should be noted that ANSI/ISA 84.00.01 and the risk-analysis techniques represent minimum standards practices and not maximum ones. The same applies to OSHA's PSM, EPA's RMP regulations, and to other industry standards. So, merely meeting the standards and regulations may not necessarily make a plant or facility safe.

What S84 Does (and Doesn't) Cover

S84 covers safety instrumented systems (SISs). Indirectly, it can put requirements on identified layers of protection. However, it doesn't have requirements of some other safety systems discussed below. It should be clear, however, that OSHA's PSM and EPA's RMP cover all associated safety and environmental systems.



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Other Safety-Related Systems?

While ANSI/ISA 84.00.01-2004 provides comprehensive coverage for SISs and some minor coverage for independent layers of protection as part of its risk analysis, what about other safety-related systems that are covered by OSHA and EPA regulations? Some examples are facility/plant/unit/section manual emergency shutdown systems (ESDs), manually activated safety systems, and orphaned safety systems or layers of protection.

Facility/plant/unit/section manual ESDs are general-purpose, operator-initiated shutdown systems, commonly known as the "Big Red Button." While these may be activated by detection of a known hazard, they're typically a more general-purpose action of last resort (AOLR) against unidentified hazards, failure of all layers of protection for a hazard, an unexpected propagation path of a known hazard that bypasses existing layers of protection, or a catastrophic event. There are no industry standards that directly cover this type of system, though some people incorrectly assume that S84 applies to them (see ANSI/ISA 84.00.01-2004 Section 1(x)). This isn't to say that some S84 principles and practices could or shouldn't be applied to this type of system, but rather that the standard isn't directly applicable.

What about other manually actuated safety systems, such as fire and gas and toxic detection systems? Some of these are automatically actuated, but many are manually actuated. Some may not even have a direct manual actuation of a safety system, and so the operator is expected to diagnose and solve the problem. These typically are actuated once the cat is out of the bag, and will hopefully minimize the result of the hazard. If these are considered safety or environmental protection systems, then they're covered as manual and operator-action systems under PSM and RMP.

What about orphaned safety systems or layers of protection? An orphaned safety system is an identified safety system or "layer of protection" that is shown not to be "required" by the risk analysis. It either currently exists or, for new systems, is a traditional safety system for that type of process. It also can be a safety system identified by the process designers or operators as desirable, or it even may be required by process design or company standards. These systems aren't covered by S84, unless they're identified as an SIS, and they're not IPLs because the risk analysis indicated they weren't required.

These safety systems and similar ones are certainly covered by PSM and RMP in the facility's mechanical

integrity system. If you've got them, then they're covered. This isn't to encourage people to take out these systems, but to identify that they need to be managed.

One should remember that existing safety systems have evolved for many reasons based on accumulated operating experience (sort of a plant community's collective memory), and a spiffy engineering risk analysis doesn't necessarily negate the need for such systems. Existing systems are based on experience (what has happened over time and what plant people worry about), while a risk analysis is based on personnel experience, engineering expertise and analysis (what has happened in the experience of the analysts present at the risk analysis, and what may happen, according to analysis and speculation based on engineering principles). Some companies designate these as SIL "A" or SIL "0," and have specific requirements for these systems that use parts of S84 to meet PSM and RMP regulations.

All of these systems can be classified as functional safety systems (FSSs) that must be managed to insure that their integrity is maintained, and that they meet appropriate functional safety requirements, regulations and industry practices.

Functional Safety Management

Management of the SIS, identified layers of protection, and other identified safety systems that are engineered or are administrative protections come under the guise of the management of functional safety, and so are covered by requirements of OSHA's PSM and EPA's RMP as part of the management of process safety and the environment. ANSI/ISA 84.00.01 provides detailed guidance for SISs, but there is little the detailed guidance for managing other FSSs. Figure 1 provides an overview of the management of functional safety.

Functional Safety in a Plant

Managing anything in a plant is sometimes difficult due to vested interests and politics, and management of functional safety is no different. This task is often given to a poor plant engineer with no stroke to get it accomplished with few resources. To successfully manage the functional safety process, the person responsible should report directly to the facility manager. There also needs to be an outside independent auditing authority. Each plant in a facility should have someone responsible for its functional safety systems, and that person should have a dotted-line connection to the facility-responsible authority.

Management andapplication of functional safety should be consistent within the facility, in the company, and in the larger industry.

Management and application of functional safety should be consistent within the facility, in the company and in the larger industry. Consistency comes from standardization and uniform application of the standards and techniques used, and from knowledge of industry practices. Engineers must know standards, but they also must know how similar units and facilities operate. Benchmarking of common process risk isn't generally available. This can be developed internally, and should be to some extent. The common sources of industry practice are third-party consultants with plant-level and industry experience.

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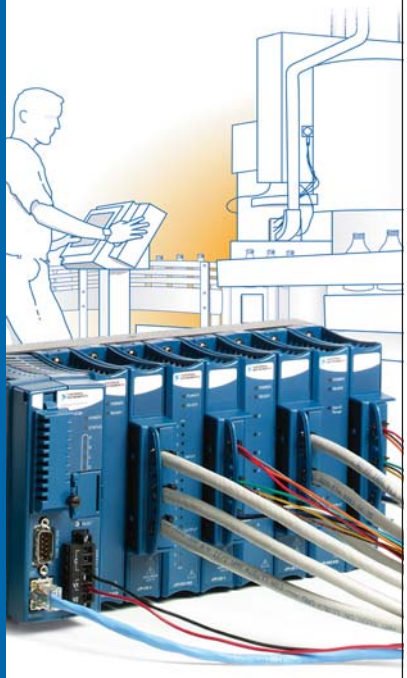

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Functional safety-system competency is another area that should be addressed. S84 standard covers it in Section 5.2.2.2. It's clear that management, implementation, and long-term support of functional safety requires competent personnel, and not just the next available engineer or employee.

SISs and other IPLs are mostly downstream protections. In short, the cat is almost out of the bag, and we're trying to keep it from getting loose. Preventing the cat from even trying to get out of the bag by reducing the demand rate on the IPLs is an alternate but complementary approach. It certainly seems reasonable that initiating causes and process design for high-risk hazards should receive additional consideration in design, operation, and maintenance to minimize safety demands. This is analogous to a quality system where the IPLs are similar to the downstream measurement of quality, but the real success of a quality system is preventing the quality defect to begin with.

Intrinsic safe design of the process is a recognized approach to this need. It's applicable to new units and projects, but is somewhat more limited for existing units. Reducing frequency of initiating causes of hazards (failures), consideration of human factors and systemic propagation of accidents, alarm rationalization, simplification (effective minimization of complexity), and a deeper consideration of abnormal conditions in process control design are also ways to reduce risk.

The Future

Some areas that need further work are standards or guidance covering the aspects of all functional safety systems, upfront reduction in risk, and a good look at process accident propagation. Accidents can consists of initiators, downstream events, human factors and systemic effects, including manpower reduction, reduction in experience levels, older plants, cultural effects and management policies. Work is also needed on risk benchmarking, overall evaluation of risk, and downstream analysis of the effectiveness of the S84 standard, its implementation, and related methodologies in reducing accidents or near misses.

There also must be an overall functional safety management plan to manage all aspects of functional safety in a facility or company. In addition, implementing such a plan requires the managing authority to have the clout and resources to enforce its management plan. It's not sufficient to let every plant in a facility manage its own functional safety, to make it a project-team responsibility, to assign it to a low-level engineer, or let it be subject to corporate guidelines with lax enforcement and implementation. The first case is the fox guarding the hen house; the second is a guard dog that's only interested as long as it's being fed; the third is equivalent to giving the task to a guard dog that is half the size of the fox; and the fourth is no guard dog at all.

All management of functional safety procedures and their implementation and operation should be audited for compliance by a third party. This should be either an independent, competent authority external to the facility but within the same company, or an outside consultant with the appropriate, plant-level experience and competence in the areas of managing functional safety. This level of auditing will help assure that apathy, political and/or incestuous relationships in a facility don't defeat the purpose of managing functional safety.

Though it's a good start, it's clear that ANSI/ISA 84.00.01 is just the tip of the iceberg toward management of functional safety. There still is much more to be done.

William L. (Bill) Mostia Jr. PE, principal of WLM Engineering Co., has more than 25 years experience applying safety, instrumentation and control systems in process facilities. He can be reached at wlmostia@msn.com.

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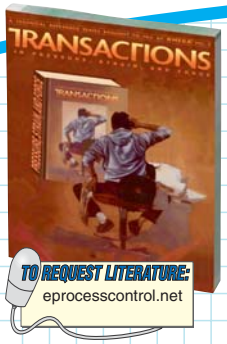

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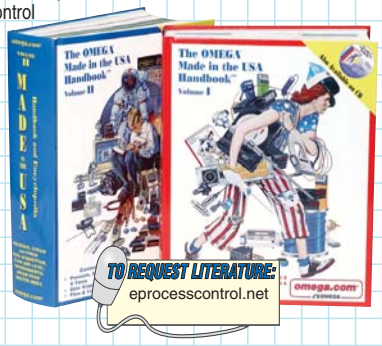

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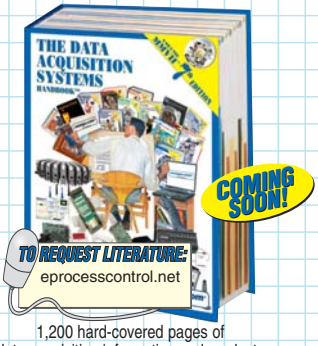

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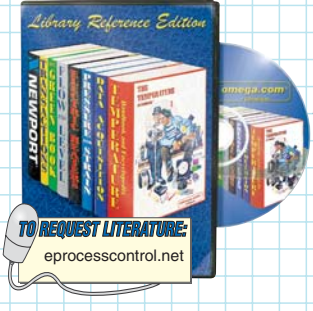

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THE MOD SQUAD: PROCESS AUTOMATION AT DOW

What, Why and How Dow Chemical Does Process Automation Differently Than Everyone Else

by Margaret Walker, Ed Sederlund, Jerry Gipson and Eric Cosman, Dow Chemical Co.

More than 30 years ago, the Dow Chemical Co. embarked on what ultimately evolved into a global corporate initiative: to develop and broadly leverage a standardized, highly integrated process automation system, incorporating basic process control, process information and safety-system functionality. Based on what we wanted to accomplish, and a lack of commercial offerings at that time, we developed several proprietary solutions, including a process control system that came to be known as MOD, which is short for "Manufacturing Operating Discipline." This article describes our journey to develop our own process control system, discusses lessons learned, and explains why we took the path we did.

Founded in 1897 by Herbert H. Dow in Midland, Mich., the Dow Chemical Co. is a global leader in science and technology, providing innovative chemical, plastic and agricultural products and services to customers in 175 countries. From the very beginning, continuous improvement has been at the heart of the company, and forms the essence of its mission: "To constantly improve what is essential to human progress by mastering science and technology." To achieve this, we're committed to the principles of sustainable development. While it might sound lofty in some circles, this underlying philosophy has been an instrumental driver in our process control journey from the beginning.

FIGURE 1.

LEVI LEATHERS

Post Street Archives

Joel Franklin Monroe (Levi) Leathers, a research and development pioneer, who became Dow's executive VP, and planted the seeds for its Manufacturing Operating Discipline (MOD) series process automation system.

The roots of our process automation philosophy go back to the 1960s, and are based on our operational strategy. Consistency and sustainability are key elements of that strategy. As a global company, it's important to operate facilities in the same way, every time, to achieve consistently high quality, as well as process reliability and repeatability of best practices. We were striving for true "operational excellence" many years before it was coined by industry analysts.

Operational excellence includes safety performance, which was and is a key focus for Dow. We're passionate about it. Protecting people, the community and the environment are non-negotiable priorities in our corporate culture. We must operate our plants safely, and in the same way, every time. Our "Vision of Zero" translates to zero incidents, injuries, illnesses, accidents and zero environmental harm. So our safety practices need to be repeatable and re-deployable as well.

We're also driven to develop our own solutions by our engineering culture. We're encouraged to look at problems and solutions differently. We ultimately integrate design, control and safety as a standardized system because we need solutions that improve the process. We design the solution around the process, not the process around the solution. Which element drives the development process is an important distinction here. You size the shoe to fit the foot. You don't change the size of the foot to fit into the shoe. So, the needs of the process and our ingrained operating discipline are what ultimately create the solution. Our Manufacturing Operating Discipline (MOD) series process automation system was designed not by automation engineers, but by chemical engineers, who were intimately familiar with the intricacies of the process.

This was the challenge, and we absolutely needed to accomplish all of it. However, in the beginning, commercially available process automation systems didn't give us the capability that we needed. Technology was available that could be applied to the problem, and we developed

FIGURE 2.

OOOOH! SHINY!

With enough idiot lights and pushbuttons to satisfy any primate, Dow's first attempts to automate were complex and wonderful.

several solutions based on general purpose computers and operating systems. Over time, we were able to develop a system that met our needs. In fact, Dow was the first company to certify a process control system with logical separation of the safety and automation functions in a common hardware platform. However, we didn't exactly plan it that way. The driving force behind this was our passion for process safety, our operating discipline and the need to simplify the automation platform. We weren't looking to invent something revolutionary. A confluence of process automation deficiencies in the market and the right people with the skills, needs and management support set the stage for Dow's proprietary automation system.

In the late 1960s, all the plants we were familiar with operated with individual loop instrumentation, or even manually in older plants. Early efforts at automation were frustrating and cumbersome due to hardware incompatibilities and complexities. More time was spent dealing with keeping the control system running than was spent improving plant operations. This had to change. What was needed was a standardized, re-deployable process automation solution that supported our goals of overall consistency, productivity improvements, global operating discipline and innovation, all while maintaining high safety performance. The common thread throughout was our passion for process safety and this operating discipline.

We were striving for true "operational excellence" many years before the term was coined by industry analysts.

Levi Leathers' Operating Discipline Vision

One of the greatest influences on MOD's early development was Joel Franklin Monroe (Levi) Leathers, a research and development pioneer, who became Dow's executive VP (Figure 1). Levi was a larger-than-life individual, who had a tremendously positive impact on those around him, both personally and professionally. You may be fortunate enough to encounter someone like him once in your life.

In the early 1960s, Levi was R&D director and later general manager at Dow's Texas Division. Under his direction, new processes and vast improvements to existing processes came into being. His words ring as true today as they did then: "If it doesn't work, there is a reason why. If it does work, there is a way to make it better." He embodied Dow's passion for safety and operating discipline, and planted the technological seeds that brought them together. He didn't know all that had to be done to achieve this goal, but he knew how to get people excited about the idea. The term he used to express his concept was "Operating Discipline."

Levi shared his vision by traveling around the company to Texas, Midland, Sarnia, Canada and many other Dow sites worldwide. He didn't just talk to control engineers, but to people at all levels, including process and instrument engineers, manufacturing, plant or line people, business leaders and senior management. Levi wasn't the actual technology inventor, but his influence is easily seen in the early development efforts that began during this decade.

Converging Activities

Over the years, there were technology development activities throughout Dow, each addressing a different aspect of the overall automation opportunity. These activities ranged from creating process information systems to developing closed-loop control applications. Ultimately, they all converged to one set of systems, but this took several years. Information sharing between engineering teams at multiple sites wasn't what it is today. Travel between sites was limited, and the technology that today facilitates information exchange between sites and work teams that we now take for granted wasn't a reality back then.

Our Manufacturing Operating Discipline (MOD) series process automation system was designed, not by automation engineers, but by chemical engineers, who were intimately familiar with the intricacies of the process.

MOD 1 is Born

One major activity began about 1968 in the Midland Analog Simulation Facility, which was seeking better production process control. By performing differential equations on analog computers, researchers found they could simulate any conceivable control strategy. As a result, analog simulation was widely used to perform studies that would determine the best control strategy for controlling a unit operation or an entire plant.

However, the development of analog simulation also led to some frustrating times. Existing hardware, such as panel-mounted conventional controllers, often became a barrier to implementing the best process control methods. Compromises also had to be made to fit existing hardware.

Despite these constraints, some process control ideas were implemented on industrial analog computers built to withstand hostile environments in the field. However, while they had the flexibility to implement control ideas without hardware barriers, analog computers were costly and very limited in capacity. The logical question was: how to implement the technology of a large analog computer in the field at a much lower cost to the company?

The answer was MOD 1. Started in 1969 on a batch distillation column, MOD 1 resembled an analog computer, and had similar logic and computing components. It had approximately one-quarter of the power of the present MOD 5 computer, but it was the foundation for sequencing, alarms, flexible analog and digital control—the basic ideas for automating a chemical process. Built by Taylor Instrument Co., which is now part of ABB, the first MOD 1 cost $85,000.

This Operating Discipline philosophy was implemented using the "sequencer" concept. The importance

and value of subdividing the entire plant into unit operations (sequences), and then subdividing the time up into steps for each sequence boggles the mind. Plant people could now think about, document, and automate all of the actions required to operate a plant, including start-up and shutdown. The control system would then provide relentless enforcement of the best way to run the processes.

These ideas often met with opposition because they were viewed as impractical or unachievable. The trench-by-trench effort to get the plants to accept and use this level of automation was fought for many years, and is still being fought in some instances. However, the favorable economic impact of the successful implementations were felt throughout our company.

MOD 2 Moves to Modular Mode

By 1970, MOD 2 was created to reduce costs, and move from a standalone to a mode-modular system. It was the first process control system with a standard system packaging method. MOD 2 also used digital modules from Digital Equipment Corp. and hardwired logic cards. However, MOD 2's analog function modules were designed internally.

MOD 3 Increases Control Capability

Coming into general use in 1972, MOD 3 was developed to increase system control capability, and reduce overall costs. Some rewiring was required to change the control logic. Installation of a MOD 3 in a Latex plant in Sarnia, Ontario, was the first use of MOD's concepts outside the Michigan division.

MOD 3 used the first all-digital CPU, which was called the Alpha 16 or the Naked-Mini. This CPU was made by Computer Automation Inc., and had 4 KB to 32 KB of memory. MOD 3 also was the first process control system to use DOWTRAN, a computer language for end-user programming.

MOD 4 Goes Global

MOD 4 followed in 1975. It was the first process control system to perform all temperature and flow calculations in its software. MOD 4 expanded the I/O capability of previous MODs, and displayed values in engineering units, instead of percentage of full scale operation. MOD 4's LSI-2 CPU expanded memory to 80 KB, and was again built by Computer Automation. Mod 4 finally made it possible to backup all inputs and outputs. Installation of a MOD 4 in the Methocel plant in Stade, Germany, was the first use of MOD systems outside of North America.

MOD 5 Brings the Best Together

Dow began using MOD 5 in 1980. Most of the first process control concepts remained in the MOD 5, but were greatly enhanced by the accumulated knowledge from implementing the system worldwide. MOD 5 represents hundreds of person-years of experience for automating and controlling processes accumulated over a mere 16 calendar years.

MOD 5 also represents the true collaboration of Dow's multi-discipline work teams (chemical engineers, process automation, instrumentation, etc.). The MOD series development was a pivotal endeavor that linked our technology center philosophies with our operating discipline. We understand better that cooperation between research (get the right process), engineering (use the right equipment), maintenance (fix it once, never again), production (optimized and safe operations), business (automated scheduling, shipping, etc.) and many other groups must occur and improve for our company to survive and excel.

We really started to see tangible, operational benefits with MOD 5, including repeatability, as well as consistency and productivity improvements. Repeatability of best practices was extremely important because it enabled us to operate our plants safely, day in and day out, no matter what else we

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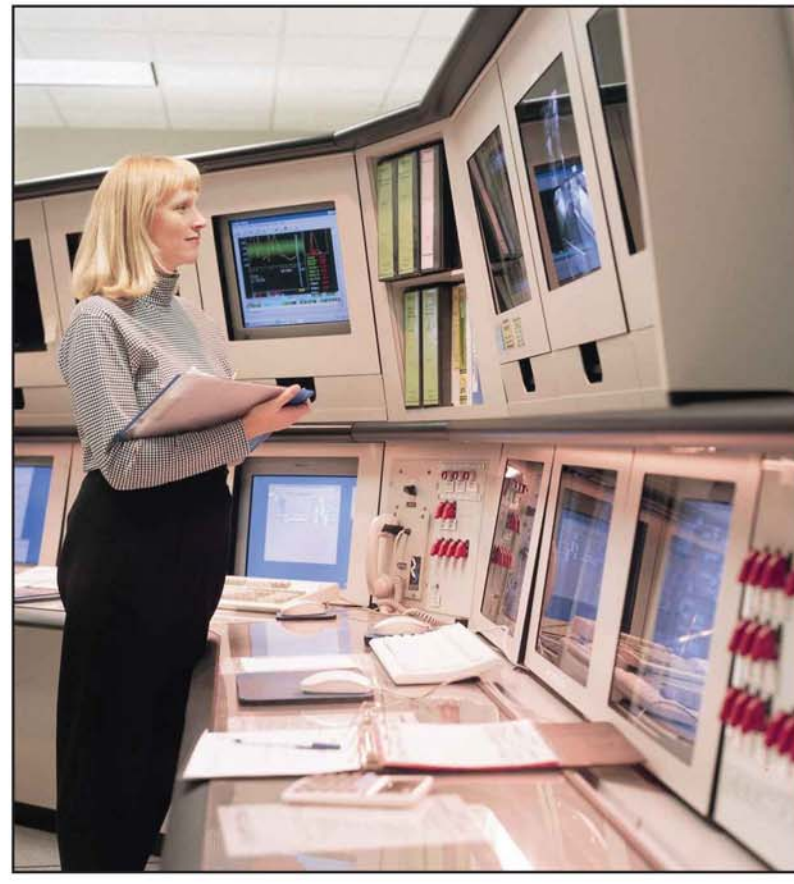
FIGURE 3.

THE MOD SQUAD GETS DISPLAYS



The MOD Operator Station provides a standard graphical interface to the process control system as well as an integrated environment for development.

were doing. It helps us to do it the same way, every time.

Consequently, in the mid-1980s, Dow decided to implement MOD 5 globally. The standard systems approach used here was extremely powerful because it means anyone familiar with a MOD 5 system can immediately relate to the MOD 5 hardware and software running a process anywhere in the world. As of 2000, we had 1,500 systems installed and more than 400 information systems in place.

In the 1990s, there were several enhancements to MOD 5. A control computer was provided that physically combined control and safety functions in the same hardware platform. However, these functions were logically separated. This version of the system was SIL 3 certified. Dow was the first company to certify a process automation system with logical separation of the control and safety functions.

There were also major projects to address the human interface to the control system and the information management functions. The MOD Operator Station provides a standard graphical interface to the process control system, as well as an integrated environment for control programming development. A single standard information system and process historian was created by combining technology from several products developed previously.

Significant Results

By combining our best technology, we've seen tremendous productivity benefits added to fulfilling our operat-

Embedding our operating discipline concept ... also helped us achieve our environmental, health and safety goals. All of these performance improvements contribute hundreds of millions of dollars to our businesses.

ing discipline and safety performance goals. We reduced control strategy development time by 45%, and reduced support costs by 50%. Process startups are automated and coordinated across multiple unit operations. Alarms are "intelligent" in the sense that they reflect the state of the process, rather than just a deviation from a limit. Batch processes are fully automated, resulting in reduced cycle times.

Embedding our operating discipline concept in the automation system also helped us achieve our environmental, health and safety (EH&S) goals. All these performance improvements contribute hundreds of millions of dollars to our businesses.

Looking Ahead: New Sustainability Options

Of course, all strategies must adapt. So, the time came when we realized it wouldn't be cost-effective to continue to invest in proprietary hardware and software systems. In early 2000, we decided change was necessary. While we wanted to preserve our process automation expertise, we knew that we didn't want to be hardware providers, and so we began to look at other options.

The next article in this series will describe our quest to find a sustainable process automation solution that could take us into the future. MOD 5 would be a very tough act to follow.

Margaret R. Walker is vice president of engineering solutions for Dow's Technology Centers and Manufacturing and Engineering Work Process. She is responsible for all of Dow's global engineering and construction, including its Design and Construction, Process Automation, Process Engineering and Process Safety organizations. She also serves as board of governors' chairman of the Synthetic Organic Chemical Manufacturers Association (SOCMA).

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Edward R. Sederlund is Dow's Process Automation product manager, and manages its lifecycle of globally deployed commercial and proprietary process automation systems.

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The authors would like to thank Peter J. Kindt, Dow's Process Control and Advanced Control discipline leader, who also contributed to this article.

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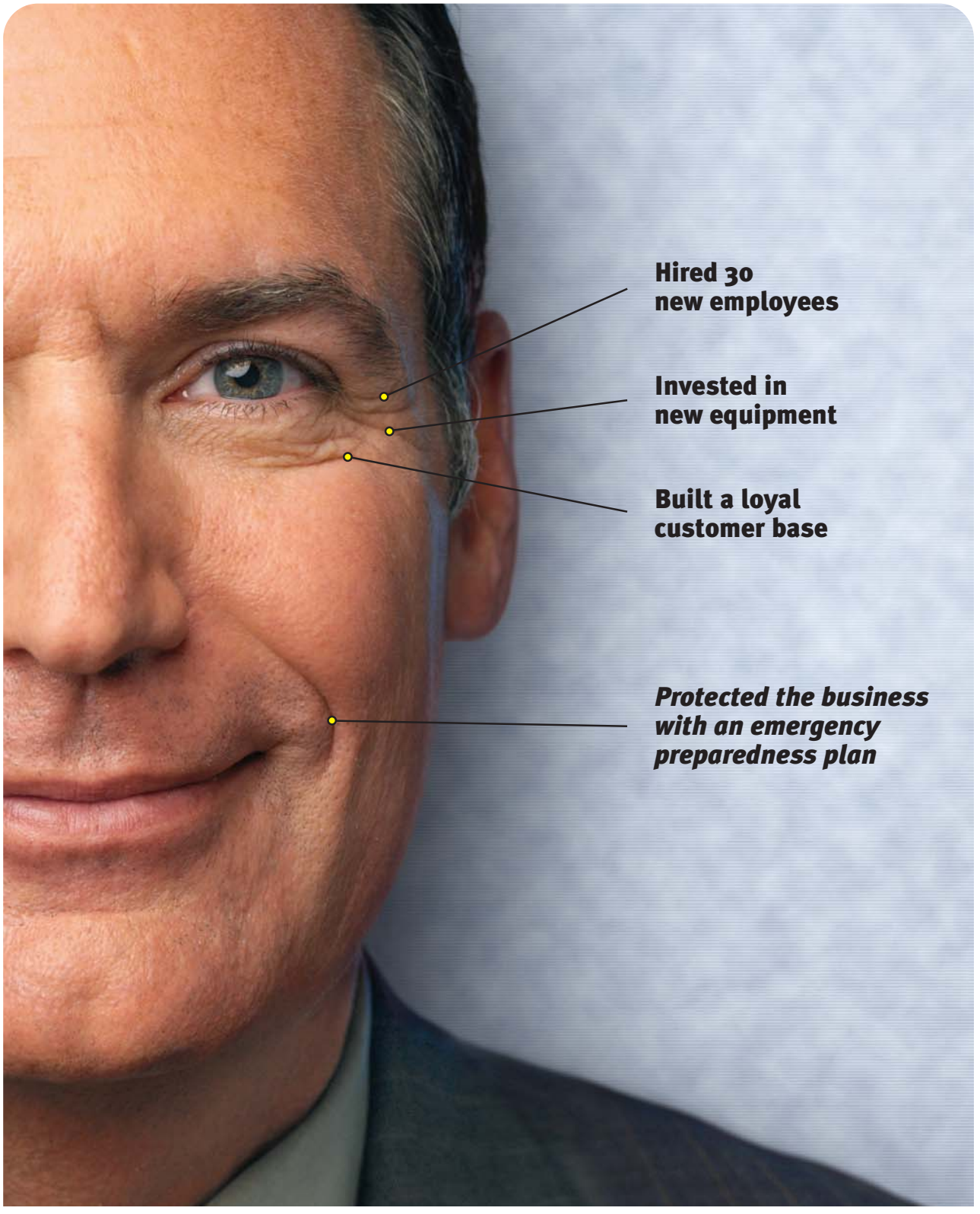

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Standards Bring Order to Chaos

Sampling and Sensor System Standardization Simplifies Design, Installation and Operation

Do your process analyzer tubing installations look like a bowl of spaghetti that someone accidentally dropped on the floor? Is every installation different, depending on which tubing and fittings were used and how they were installed?

The New Sampling/Sensor Initiative (NeSSI) will address these issues by simplifying and standardizing sample system design. NeSSI also will adapt the emerging class of lab-on-a-chip sensors to a miniature/modular smart manifold with the aim of fundamentally changing and improving process analysis.

NeSSI operates under the sponsorship and umbrella of the Center for Process Analytical Chemistry at the University of Washington in Seattle. NeSSI's membership focuses on manufacturers supplying parts and components for the initiative, as well as end-users doing in-house testing.

Meanwhile, the U.S. Food and Drug Administration's (FDA) Process Analytical Technology (PAT) initiative promotes using analytical and other process sensing systems to improve manufacturing processes in regulated industries. End users want NeSSI and PAT to induce manufacturers to produce standard sampling, sensing and analyzing systems with interchangeable components.

According to Patrick Lowery, chief engineer at Circor Tech (www.circortech.com), end user benefits from NeSSI and PAT are extensive. Standardization allows end users to select from several suppliers for various functions with mix-and-match and plug-and-play capabilities because all surface-mount components are designed with the same footprint. About 50% less system area is needed with these systems. Reduced internal volume and smaller wetted-surface areas allow shorter purge times and less probability of cross contamination from one analytical sample run to the next. There also is less propensity for corrosion and surface phenomena, such as contaminate adsorption/desorption, thanks to this reduced surface area.

In addition, there's no need for end users to employ expert tube benders or unique brackets, and the requirement for thread tape and the current multitude of fitting varieties are virtually eliminated. Sealing integrity is greatly improved, and leak event frequency is dramatically reduced. Also, better sealing yields lower maintenance costs, higher safety standards and less fugitive emissions.

Steve Doe, analytical market manager at Parker Hannifin (www.parker.com), says his firm has been actively developing advanced functionality for its NeSSI platform of IntraFlow products for more than two years. Parker's block heaters are placed in direct contact with the back side of the system backplane to conductively heat the system, as opposed to traditional convective heating arrangements.

Parker's double-block and bleed-stream switching is also available, as well as pressure and differential pressure sensors, inferring flow characteristics to provide remote feedback capabilities. "These additional functionalities provide a fully integrated pressure, temperature and flow control and monitoring package," claims Doe.

Another leading NeSSI vendor is Swagelok (www.swagelok.com). "Our MPC platform eliminates most of the tubing and fittings normally used in assembling traditional sample system panels," says Dave Simko, Swagelok's marketing resources manager. "The only tubing and fittings needed are those few connections required to install the platform into the complete process analyzer system, and to connect support systems for stream selection, calibration, purge and carrier."

Simko adds, as FDA's PAT guidance is implemented, there will be many applications for NeSSI systems. "PAT was promulgated in its final form in September 2004, and the pharmaceutical manufacturers are determining which analytical techniques and methods are most applicable to their specific processes. Some applications have already been implemented," reports Simko.

Simko says that end users already report reducing the cost of designing and building process analyzer systems by 40%, and cutting the cost of owning these systems by 35%. "The next step for end users and process analytical OEMs is to integrate these systems with process analytical devices to create a complete integrated solution," adds Lowery.

Dan Hebert, PE, Senior Technical Editor
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ControlGlobal.com presents "Ask the Experts," a column moderated by noted process control authority and columnist Béla Lipták. Save yourself hefty consulting fees by getting the answer to your question from Lipták's cadre of professional automation experts.

Question:

I have a problem controlling latex (pure rubber mixed with acids) temperature inside a batch reactor. The previous equipment was pneumatic, and has now been converted to electronic. Because our controller can't perform auto tuning, we have to tune it manually. Our customer isn't satisfied with current readings on chart paper, where oscillation is too high and over-responsive. Please advise us how to perform manual tuning for cascade control processes.

Our customer also said that control valve response is too fast. They want measurement close to the set point, with the valve responding slowly (closed or open) until both readings are the same. What happens when measurement exceeds the set point?

Zulkefli bin Zakba,
project engineer, Foxboro (M) Sdn Bhd

Answer:

Prevent Limit Cycling

I will try to give you a brief summary but the tuning I will describe is only for the exothermic (cooling phase) of the batch reaction.

From the brief description you gave of your system's present behavior, a simple cause of the cycling could be the dead band of your steam valve, which can be caused by dirt buildup on the valve stem. This can be fixed by cleaning the stem, or by placing one on the valve if you don't have a positioner. The positioner should have a gain of 20 or more (PB = 4-5%), and the actuator should be strong enough to provide it.

"Our customer isn't satisfied with current readings on chart paper where oscillation is too high and over-responsive."

Another cause of limit cycling can be an interlock that prevents the opening of the steam valve after the initiator or catalyst is added to start the reaction. This can cause a limit cycle because the addition of the initiator can cause a drop in temperature. If the master TRC is prevented from correcting, its integral will wind up, and could cause a limit cycle. It is one reason why we always provide external reset (Figure 1).

If your cycling is not caused by the above reasons, it could be caused by the hierarchy of speeds of the three loops of your cascade system. It's essential for good operation that the speed (dead time plus time constant) of the inner loops be much faster (if possible an order of magnitude) than their outer loops. This means that the valve must respond faster than the slave TRC, and the slave TRC must be much faster than the master TRC.

If none of the above is the case, then start looking at retuning the master.

Tuning the Master

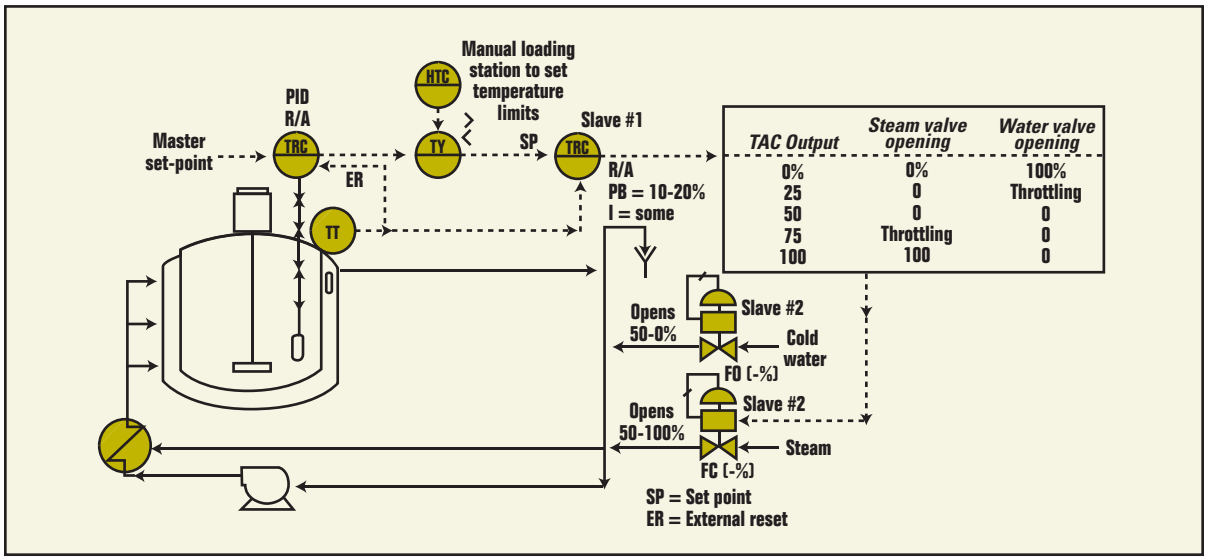
I will concentrate on tuning the TRC master, and will treat the slave TRC as if (from the master's perspective) it was part of the control valve. When visiting plants, I only check the positioner gain, which must be around 20-25 (proportional band of 4 or 5%), and set the gain of the slave TRC to ¼ of that (gain = 5-6 or PB = 15-20%).

If the slave measurement is noisy, I do not use the "I" mode at all, and usually don't use the "D" mode either (if the measurement is noisy or if the loop speed is fast enough as is). If neither is the case, I put in a few seconds of derivative, but only on the measurement, and not on the set point.

Now, to control the cooling phase of the batch process, we have to match the dynamics of the master PID controller to the dynamics of the process. The dynamics of any process can be described by at least three parameters: gain (Kp), time constant (T), and dead time (Td).

FIGURE 1.

ALLOW TRC TO CORRECT



If the master TRC is prevented from correcting, its integral will wind up, and could cause a limit cycle.

Open-Loop Tuning of Non-Interacting PIDs

The process gain (Kp = K/?m) in your case is the % temperature change resulting from a 1% change in valve opening (coolant flow). In case of a chemical reactor, this gain drops as the load (reaction rate) increases because the heat-transfer area of the reactor jacket is fixed, and, therefore, the efficiency of heat transfer drops (Kp drops) as the load rises. You want to compensate for this drop in Kp by using an equal percentage valve (= %), which increases its gain (Kv) as the load rises. This way, the KpKv product stays relatively constant.

The PID control loop gain in the product consists of four gains: process gain (Kp), valve gain (Kv), sensor gain (Ks), and gain-setting of the controller (Kc, which is the P in the PID). We usually tune the gain (proportional band) of the controller (Kc) to obtain ¼-amplitude damping of the loop's response to a load upset. This means that after an upset, we want (after each cycle) the amplitude of the oscillation to decay to ¼ of the amplitude in the previous cycle. To obtain this rate of decay, the product of the four gains (Kp.Kv.Ks.Kc) has to equal 0.5.

In open loop tuning of self regulating processes, when the reactor is operating in a steady state, the PID master of the cascade loop is switched to manual, and a step change is manually introduced into its output, manually reducing (or increasing) the coolant valve opening. This step-increase or reduction in the heat input results in a reaction curve (a drop or rise in reactor temperature), which describes the dynamics of the process. The steady-state gain of the process is the ratio of the % increase (or decrease) in temperature per % decrease (or increase) in coolant flow (K/?m). The time it takes for the reaction curve to reach 63% of its final value (0.63K) is the sum of the dead time (Td) and time constant (T) of the process. The split between these times is obtained by the intersection of a tangent drawn to the inflection point of the curve. The slope of this tangent is the reaction rate (A/T). Based on these readings, the Ziegler-Nichols' recommendation for the correct settings can be calculated.

There are several reasons why open-loop tuning is seldom used on batch reactor control. One is that it requires switching the controller to manual; another is that it upsets the process. If you're not sure if the process is self-regulating, you shouldn't try this method due to the risk of causing a runaway reaction or stopping the reaction. To reduce the size of the upset, operators tend to make the step change small. However, a small step usually results in a small response, which (particularly if the process is noisy) can be hard to read by determining where the 0.63K or the inflection point is and what the slope of the tangent at the inflection point should be.

[*Editor's note: For an expanded version of this article, view it online at www.controlglobal.com*]

Béla Lipták,
CONTROL columnist
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HMI Software is Disappearing

HMI software is getting better and cheaper, but it's disappearing into control system vendors' portfolios.

As we pointed out last year in this space, providing information to an operator is fast becoming a human-machine interface's smallest task. That an HMI can provide trend and alarm displays is a given: all HMIs do it and in just about the same way. Walk around at an ISA Expo, and you see the same displays from booth to booth. HMIs that just do operator interface tasks are a commodity, and you can buy them dirt cheap off the Internet.

The real action is in HMIs that provide web access, interface to higher-level enterprise software, perform MES functions, support Windows CE, and interface to every type of control system. The problem is there aren't many HMI packages that can do all that, and they're also pulled off the market on a regular basis. For example, Invensys bought Wonderware, GE Fanuc bought Intellution, and Schneider Electric most recently bought Citect.

Truth be told, many software packages on the market are re-branded versions of products from Wonderware, Intellution, Citect, Iconics, InduSoft and one or two others. With three of those companies now in the hands of process control vendors, many companies using branded versions of those packages are getting nervous. Do they really want their HMI packages controlled by those three big companies? If not, it means they either have to develop HMI/SCADA packages themselves, find another supplier to re-brand, or buy up one of the few remaining independent HMI/SCADA vendors.

It's not likely that equipment vendors will develop their own because HMI/SCADA software is a tough business. Not only is it extremely price competitive, but doing all the new functions associated with a modern HMI is very tricky, requiring very specialized expertise in Windows operating systems, enterprise integration and web technologies. It's easier to buy these capabilities.

For example, the Gartner Group (www.gartner.com) says the reason Schneider bought Citect was to obtain Citect's MES capabilities. One observer says the reason GE Fanuc bought Intellution was to get its process historian capability. Some major vendors are seeking HMI software with Windows CE capability.

To get an idea of how widespread certain packages are through the industry, search an HMI/SCADA vendor's web site for news of "deals" between the vendor and equipment makers. For example, search Indusoft's site (www.indusoft.com), and you'll find releases announcing rebranding and distribution arrangements with Automation Control Products, Advantech, Altersys, Applied Data, Arista, Beckhoff, CCS, Fluke, HMW, Maple Systems, Nematron, Optimation, Siemens, SoftPLC, Software Toolbox, Woodhead, Xycom and others. On the Iconics site (www.iconics.com) you'll find similar arrangements with Endress+Hauser, Autosol, DVT, Saflink and others. You can bet that these announcements are just the tips of the icebergs in both companies, since many equipment vendors don't want to tell you where they get their stuff.

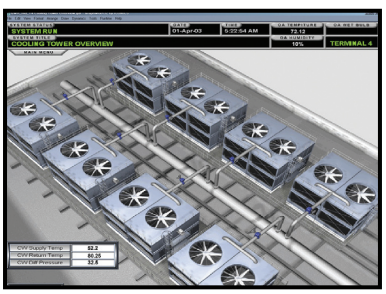
As for the cost of HMI/SCADA, it is dropping like a rock. At Automationdirect.com, for example, you can buy HMI software for prices ranging from $199 to $1,295. In the roundup that follows, you'll see an HMI PC with embedded HMI/SCADA software for $739. Small wonder the "big boys" may not want to develop their own HMI software when quality products are available at such prices.

Maybe next year at this time, all the remaining independent HMI/SCADA vendors will have been snapped up by process control and automation vendors, and we'll have a completely different story to talk about. Such as: Why has the cost of HMI/SCADA software quadrupled in the past year?

—Rich Merritt, Senior Technical Editor

HMI/SCADA MINES DATA

Enhancements to Genesis32 V9 includes support for SNMP, OPC UA and data-mining, all of which enables viewing and reporting of real-time or enterprise data sources, including Microsoft SQL Server, Oracle, plant historians, and OPC real-time data. Real-time manufacturing and business information can be viewed on a web-enabled visualization dashboard. More info at controlglobal.com. **Iconics; 508/543-8600 Ext. 202; www.iconics.com**

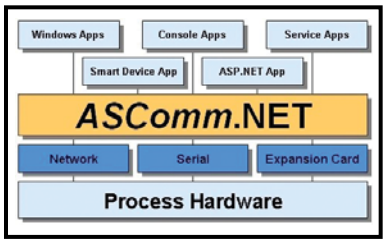


HMI REACHES PLCS

Enhancements to iFIX 4.0 include the ability to discover PLC information from Siemens and Rockwell systems; integration with Proficy, including configuration, tags, historian, real-time information portal and change management; and graphic tools for drawing 3-D piping and connected object management. More info at controlglobal.com. **GE Fanuc; 800/433-2682; www.gefanuc.com**

HMI/SCADA GETS DOT NETTED

Enhancements to ASComm.NET includes .NET communications driver and library for support of major industrial brands and protocols such as A-B, Modbus, GE and Automation Direct. The.NET library provides a 'pluggable' driver approach that allows users to add one or more drivers without affect-



ing existing code or requiring special handling. More info at controlglobal.com. **Automated Solutions; 707/575-9631; www.automatedsolutions.com**

CE TOUCHSCREEN COMPUTER

Ivory Series open platform CE touchscreen operator interface computer comes with Indusoft HMI/SCADA software. It has Internet and network access with 10BaseT Ethernet, device-ready USB ports, compact flash port, and RS-232 and RS-485 com-

patible com ports. Models include a 6-in., 256-color STN touchscreen computer. More info at controlglobal.com. **Maple Systems; 425/745-3229; www.maple-systems.com**

PROGRAMMABLE, TOUCHABLE PANEL

C-more touch panels are fully programmable, so objects can be placed, scaled and overlapped with-

out limitation. Panel displays are available in either grayscale or full color. STN grayscale and color displays are available in a 6-in. touchscreen, while color TFT touchscreens are available in 6 to 15-in. versions. More info at controlglobal.com. **AutomationDirect; 800/633-0405; www.automationdirect.com**

HMIS FOR LABVIEW

FPT-1015 flat-panel touchscreen, TPC-2006 touch-panel computer, and PPC-2015 panel PC HMIs work with LabView graphical development environment to develop user interfaces. Engineers can deploy LabView applications on HMIs and connect via Ethernet to other hardware through industry standards including Ethernet, USB, PCI and PCMCIA. More info at controlglobal.com. **National Instruments; 512/683-5090; www.ni.com**

VB SCRIPTING FOR WEB VIEWING

Enhancements to Web Studio 6.1 HMI/SCADA software include Visual Basic scripting, a trending tool,

screen translations, an advanced development environment, and database management functions. An optional wizard imports legacy PanelMate applications. VB scripting allows HMI screens and applications to be ported to Windows CE devices or viewed on web browsers. The software runs on all Windows platforms, including diskless CE. **InduSoft; 877/463-8763; www.indusoft.com**

TOUCHSCREEN HMI

Qterm-Z60 graphic operator interface terminal has touchscreen graphics, one switch-selectable serial port,

and can communicate via EIA-232, EIA-422 or EIA-485. It has a QVGA, 320 x 240-pixel, 256-color, STN color LCD display that is viewable in most lighting conditions; 2 MB flash/16 MB SDRAM; programmable speaker; and a 5 VDC 200 mA output from the interface connector to power external devices. **QSI; 80¼66-8770; www.qsicorp.com**

EXTREME LIGHTING DISPLAYS

High Bright monitors are suitable for use outdoors or in brightly lit rooms. Displays have a screen luminance of 700 nits with a high-contrast ratio, and provide VGA to XGA screen resolutions. Monitors are available with resistive or capacitive touchscreens in 15 and 17 in. models, and in NEMA 4 and 4X (IP65/IP66) industrial panels and enclosures. **VarTech Systems; 800/223-8050; www.vartechsystems.com**

Level Indicators Reach New Heights

Users say Magnetrol's Eclipse and Orion's Aurora technologies show them what's going on

by Jim Montague, executive editor

Exactly how much is in there? It's an easy question if you can see into a tank, but what if you're basically blind, or your tanks must be sealed? Sure, traditional solutions may work for awhile, but they're historically hard to maintain and sometimes lack accuracy.

Some users are implementing Eclipse guided-wave radar (GWR) and Orion's Aurora magnetically coupled level indication technologies from Magnetrol International (www.magnetrol.com) and Orion Instruments (www.orioninstruments.com).

GWR is based on time-domain reflectometry (TDR), in which pulses of electromagnetic energy are transmitted down a probe, reflected when they reach a liquid surface, and measured to indicate level. Aurora combines the operating system of a float-based, magnetic-level indicator (MLI) with Eclipse's technology. Change in the process tank level corresponds to change within Aurora's chamber. The float in the chamber actuates flags or a shuttle for visual indication, while its Eclipse GWR transmitter provides continuous electronic level measurement.

Tim Lingg, senior process engineer at Oxy (formerly Vulcan Chemicals) in Wichita, Kan., says his facility most recently installed Aurora units on its liquid chlorine processing vessels. He's also had Eclipse devices on chlorine storage and in a 50% caustic plant for about four years. These replaced elderly differential pressure level transmitters.

"We had to use nitrogen purges through the differential pressure meters, but it was hard to keep them clean, and we still saw a lot of plugging." says Lingg. "So, when we put in Eclipse, we put in two transmitters per storage tank on each of five 50-ton tanks to make sure they'd agree with each other, and they do.

"This is a much more hardy solution, and we now have a lot more time between maintenance tasks. We're thinking we may be able to go at least two or three years between cleanings," Lingg concludes.

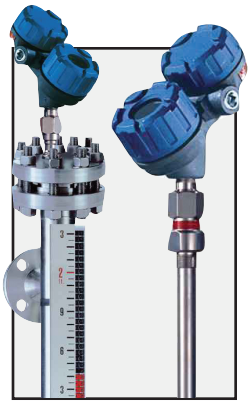
The fact that Aurora incorporates Eclipse in it allows both local, visual level checks, and electronic checks that are sent back to Oxy's DCS. Lingg adds the beauty of Aurora is that its indicator isn't glass, but rather a magnetic float inside the vessel, which actuates the indicator on the outside. This means Oxy's staff no longer has to clean iced-up indicators by blasting them with steam.

Similarly, Louis Tafoya, control analyst at Chevron's Salt Lake City refinery, says he presently uses Eclipse indicators to measure level on four vessels and one sulfur pit.

"Eclipse is trouble-free, rugged and reliable. The guided-wave radar goes through the solid tube, hits the medium, and reflects back," says Tafoya. "This means we can't overfill or over pressure our vessels, which would cause upsets in the system. So, as new applications come up, we're probably going to look at using Eclipse more."

Though less hazardous then petroleum and chemical processes, biotech application often have to be far cleaner. Steve Ward, co-owner of AQC Calibrations, in Discovery Bay, Calif., says these processes initially used other mechanical devices, but they were hard to set up, prone to errors, and typically in sealed vessels that couldn't be opened once their processes had started.

FIGURE 1.
AURORA EMERGES



Aurora combines a float-based MLI with Eclipse.

"We had to come up with a way to replace level transmitters once these processes were underway, and be able to calibrate transmitters without pulling the probe out and opening the tank," says Ward. "These tasks used to mean a half-day shutdown, so we worked to refine these processes to make them repeatable and have validated calibrations."

Ward adds the 21-ft probe in these fermentation vessels has targets set on it, which allow readings to be taken using a Leica laser. Once the level probe was set, a tank could be filled via a MicroMotion flowmeter, and measurements could be applied to a strapping table, which Ward adds Magnetrol has since incorporated into its transmitter to linearize inputs and outputs to PLCs.

"Now anytime we have a tank in a clean or hold state, we can remove the calibration transmitter to a test bench, and run a calibration in half an hour, instead of half a day," says Ward. "Magnetrol's response has always been to make the improvements we've requested. Over the years, they've responded tremendously as we've redesigned these sanitary systems, and that's made these technologies very solid."

Wonderware's Single Node ArchestrA

ArchestrA is an Invensys technology developed by Wonderware (www.wonderware.com) who packages the technology inside of its Industrial Application Server (IAS) development and run-time environment, InTouch View, and Industrial SQL Server historian. The current implementation of IAS supported distributed, network-based operation only.

"At the time of the version 2.1 release, we were working on the additional testing and packaging of this new single node offering, and it is now available in the product," says Mark Davidson, Vice President of Global Marketing, from Invensys Wonderware.

"In the past, ArchestrA's platform services were only intended to be used across a TCP/IP network with a distributed network of PCs," says Nancy Venable, Wonderware's VAR/OEM Program Manager.

Now the latest release of IAS 2.1, allows ArchestrA's platform services to be installed on a single PC, along with InTouch visualization and Industrial SQL Server historian, making ArchestrA technology viable for small system applications that use a single PC instead of a network. "All of the resulting applications can now be scaled up or down from a standalone node to a multi-node configuration that involves hundreds of computers and up to 1 million I/O," Venable says. "This allows smaller process and batch process users to take advantage of ArchestrA's benefits without having to install and support a network of PCs and servers," Venable continues. According to the company, these benefits include lower cost of implementation with application template libraries, and solutions that can seamlessly integrate into end users' existing systems.

Single Node ArchestrA technology is built upon open industry standards including Microsoft's .NET and SQL Server and also supports OPC, XML, SQL, OLEDB, ISA-88, and ISA-95.

Invensys Wonderware; 949/727-3200; www.wonderware.com

INTERFACE LINKS LONWORKS TO CONTROLLOGIX

SlotServer interface module connects devices using LonWorks to the ControlLogix 1756 platform. On the LonWorks side, the module can transfer up to 2,500 SNVTs between ControlLogix and LonWorks, connecting to up to 15 LonWorks devices per unit. Standard network managers such as LonMaker can be used to bind the module into the LonWorks network. On the ControlLogix side, users can control communications in ControlLogix software using standard, generic message blocks. Sample message block code is provided in an ACD file with each module. Typical LonWorks applications include energy monitors, HVAC controls, I/O controls, lighting controls, motor controls and sensors. **FieldServer Technologies; 888/509-1970; www.fieldserver.com**

SAMPLING STREAM SELECTOR

Stream selector system delivers a representative sample from multiple sample streams to one analyzer. The modular assemblies are ANSI/ISA 76.00.02 compatible and accommodate multiple process streams in limited space. Each stream is controlled by a double-block and bleed (DBB) module to eliminate cross contamination and maintain sample integrity. More info at controlglobal.com. **Swagelok; 440/349-5934; www.swagelok.com**

CLAMP ON AND GO ON

DFX clamp-on Doppler ultrasonic flowmeter can start measuring flow in a couple of minutes without special

tools, training or specialized computer software. Intelligent software algorithms monitor incoming signal quality and flowmeter performance during initial startup and continuously during operation, making automatic adjustments to nullify changing liquid conditions. More info at controlglobal.com. **Dynasonics; 262/639-6770; www.dynasonics.com**

FOOD DATA LOGGER

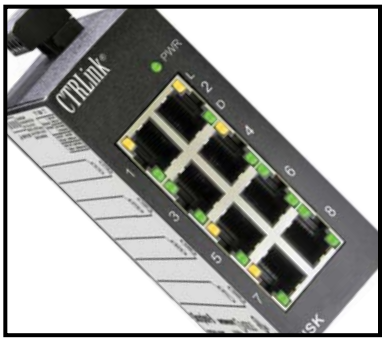
Mini Data Logger (MDL) is suitable for food process verification, pharmaceutical storage, laboratories, transportation of temperature-sensitive goods or monitoring equipment run time. Once desired temperature information is recorded, the logger can be plugged into a computer's serial port for downloading and viewing logged data. **Dwyer Instruments; 219/879-8000; www.dwyer-inst.com**

DRIVE THAT CABLE

Variable Frequency Drive (VFD) cables carry power from AC drive systems to AC motors. The cables resist problems caused by high power levels and voltage extremes that can lead to a corona discharge between the cable's conductors. They have thick, industrial-grade XLPE insulation that provides lower capacitance and stable electrical performance, and heavy-duty PVC jackets for resistance to oil and sunlight. More info at controlglobal.com. **Belden CDT Electronics; 800/BEL-DEN4; www.belden.com**

ETHERNET SWITCH NEGOTIATES

EISK8 Skorpion Ethernet switch provides eight 10/100-Mbps, shielded RJ-45 ports. Each port supports the autonegotiation protocol to select data rate, duplex operation and flow control. The switch has a 0-60 °C temperature range, redundant power connections and troubleshooting LEDs, and is in-



dustrial EMC and RoHS compliant. It can be powered from AC or DC. **Contemporary Controls; 630/963-7070, x116; www.ccontrols.com**

MORE POWER TO FIELDBUS

Route-Master Series 100 fieldbus system has an intrinsically safe, isolated power supply that provides 350 mA per segment to drive up to 16 devices. It also can power devices over cable segments up to 1,900 meters. It

works with Foundation Fieldbus and Profibus PA networks. More info at controlglobal.com. **Moore Industries-International; 818/894-7111; www.miinet.com**

SIMULATION SUPPORTS ANALYSIS

Simulation Module 2.0 extends the LabVIEW graphical development environment to deliver a seamless plat-

form for modeling, design, simulation and real-time implementation of control systems. Engineers can more easily analyze complex, nonlinear control systems using optimization, model trimming and linearization, and decrease development time. More info at controlglobal.com. **National Instruments; 800/258-7022; www.ni.com**

MOTOR CONTROLLER DOES DIAGNOSTICS

Allen-Bradley SMCFlexmotor controller has two inputs that improve process monitoring, and provide greater access

to diagnostic information to reduce maintenance costs and downtime. Its four output devices enable users to extend control capabilities, reducing the need for a separate programmable controller. The controller has an internal bypass contactor, resulting in less heat and a smaller footprint than previous models. **Rockwell Automation; 800/223.5354, Ext. 1853; www.ra.rockwell.com**

CONNECTOR AIDS SQL, BIZTALK

Enterprise Connector supports Microsoft SQL Server 2005 and BizTalk Server 2006, providing enterprise application integration (EAI) for the company's Proficy integrated software products. These include programming, control, HMI/SCADA, historian, MES and real-time information portal software products. The connector provides connectivity to business and financial software, such as ERP. **GE Fanuc Automation; 800/433-2682; www.gefanuc.com**

CALIBRATOR MEASURES PARAMETERS

726 precision, multifunction process calibrator measures and sources pro-

cess parameters including volts, mA, RTDs, thermocouples, frequency, resistance and pressure with accuracies of 0.01%. It has a transmitter error-percent calculation function, and can interpret calibration results without a calculator. The unit can store up to eight calibration results, enabling later analysis of field data. **Fluke; 888/308-5277; www.fluke.com**

ACCURATE PRESSURE TRANSMITTER

Premium Performance pressure transmitters provide span accuracy of 0.025% for spans as low as 10% of maximum span (10:1 turndown).

When calibrated for spans of 1.25% of maximum span (80:1 turndown), accuracy is 0.050% of span. These new accuracy specifications apply to both digital communications and 4-20 mA output signals. **Invensys Foxboro; 866/746-6477; www.foxboro.com/instrumentation**

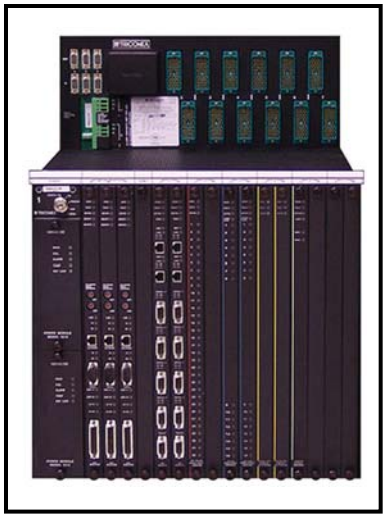
PREDICTIVE CONTROL SOFTWARE

Enhancements to DeltaV PredictPro software include handling up to 80 controlled variables and 40 manipu-

lated variables, as well as improved optimization. The software can be used for supervisory or regulatory control where multiple interactive variables are difficult to manage, such as processes having excessive deadtime, long time constants, inverse responses and loop interactions. More info at controlglobal.com. **Emerson Process Management; 512/418-7536; www.emersonprocess.com**

SIS-TO-DCS HAS OPC, TCP FOR LANS, WANS

Tricon communications module (TCM) provides safe and secure communications between safety instrumented systems (SIS) and distributed control systems (DCS). It incorporates an embedded OPC Server and supports full TCP/IP and routing functions for use with LANs and WANs. A security model enables users to specify which computers on the network can communicate with the SIS, and the type of access permitted. **Triconex; 866/746-6477; www.triconex.com**

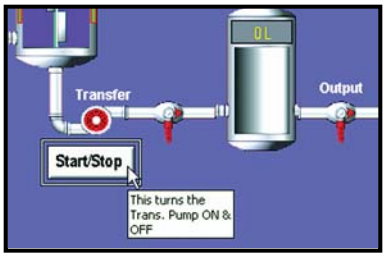


RESIST THOSE CHAINS

RP Custom resistive, chain-level measurement sensors operate with a float that travels up and down a vertical stem. A permanent magnet inside the float acts on a series of closely set reed switches and resistors inside the stem. The sensor converts the resistance value into voltage and current signals proportional to the level of the liquid. More info at controlglobal.com. **Scientific Technologies; 888/349-7098; www.stiapg.com**

DATA HISTORIAN FITS PCS

IndustrialSQL Server 9.0 Historian Personal Edition works on small applications running on Microsoft Windows XP Professional and SQL Server Personal Edition with 100 to 500 tags. The data historian can be used either standalone or added to a basic HMI system. More info at controlglobal.com. **Wonderware; 949/459-9696, ext. 244; www.wonderware.com**



SAFE GAS MONITORS

Ultima XE and XA electrochemical and catalytic combustible gas monitors, along with the Ultima XIR and XI point infrared combustible and carbon dioxide gas monitors, have a Safe Failure Fraction less than 90%, and may be used in SIL-2 applications as one device. **MSA; 724/776-8721; www.msanet.com**

IR THERMOMETER GETS ON WEB

Solonet infrared thermometer can be connected via the Internet or used as a four-wire transmitter. It has an integral web browser and Ethernet capability, so thermometers can be set up remotely from a PC without needing specialty software, and can be connected and monitored on a company-wide network. More info at controlglobal.com. **Land Instruments International; 215/504-8000; www.landinstruments.net**

MODULE CONNECTS FF TO ANY HOST

Model 3420 fieldbus interface module connects Foundation Fieldbus devices to any process control system via Modbus RTU (RS485), Modbus TCP/IP (Ethernet), HSE, OPC and wireless interfaces. This allows access to predictive diagnostic information from the fieldbus devices, enabling asset management and performance optimization. More info at controlglobal.com. **Emerson Process Management/Rosemount; 952/949-5165; www.emersonprocess.com**

IS VIBRATION TRANSMITTERS

This unnamed line of 4-20 mA vibration transmitters comply with ATEX and CSA North American intrinsically safe standards. The transmitters can be used to monitor rotating machinery in refineries, grain processing facilities, or mines, where hazardous gas or dust may be present. These 4-20 mA transmitters are loop powered, and can be interfaced with existing plant monitoring equipment such as PLC, DCS or SCADA systems. **PCB Piezotronics; 800/959-4464; www.imi-sensors.com**

PANEL WITH DETACHABLE MONITOR

ARP-1720AP industrial panel computer has a 20.1-in. detachable, NEMA 4 LCD monitor for easy maintenance and quick upgrades. The unit is available in P3 or P4 CPU versions with 24 VDC or 100-230 VAC input power, expansion slots and a RAID-1 system. More info at controlglobal.com. **Arista; 510/226-1800; www.aristaipc.com**

UNMOVED FLOWMETER DOES DIRTY WORK

With no moving parts to wear or foul, Signet 2551 insertion-style, electromagnetic flow sensor measures flow in fluids containing a high content of particles, solids and fibers, making it suitable for applications with dirty fluids. It has a built-in display, empty pipe detection and bi-directional flow capabilities. More info at controlglobal.com. **George Fischer; 800/854-4090; www.us.piping.georgefischer.com**

OPERATOR PANEL INTEGRATES I/O

PP15 Power Panel integrates I/O for control, positioning and visualization tasks. The display is IP65 protected on the front, has 4x20 character res-

olution and 26 keys, each equipped with LEDs. Two models are available; one has 16 digital I/O, while the other has four analog inputs and outputs plus eight digital I/O. Both have a CAN interface for for remote I/O and servo drives. **B&R Industrial Automation; 770/772-0400; www.br-automation.com**

CONTROL, VISUALIZATION IN ONE BOX

EZTouch integrates control and visualization for small to medium machines that require touch-type graphical displays. The system integrates a micro modular PLC base with either 32 I/O (for 6-in. units) or 64 I/O (for 8, 10 and 15-in. units) available. Snap-in EZ I/O modules can be chosen from numerous options, including analog I/O, high-speed counter, relay and AC/DC combo modules. Connectivity to Ethernet TCP/IP, Profibus and Devicenet allows users to address remote I/O, and connect to just about any device on the plant floor. **AVG; 563/359-7501; ww.avg.net**

VERY VISIBLE, PROGRAMMABLE TERMINAL

ScadaPack Vision50 terminal has a high-visibility LED backlit LCD display with four lines x 40 characters, 20

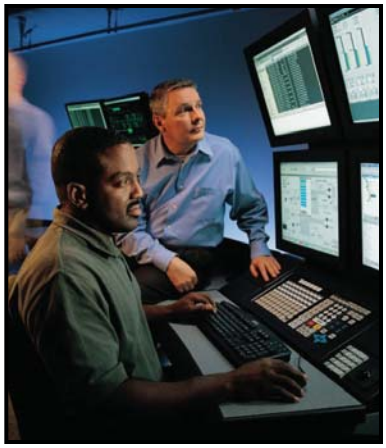
programmable function keys, and eight programmable LED indicators. Programming is via Windows-based software that provides prompts to change data and presets, and defines up to 32 alarms. It has a Class 1, Div 2 Hazardous Area Rating. **Control Microsystems; 613/591-3878 ext. 223; www.controlmicrosystems.com**

CLONING BACKS UP SYSTEMS

VersaView cloning utility provides advanced backup, data protection and recovery for users. It uses Symantec Ghost 8.0 cloning program to create a duplicate image of the computer's hard drive, so the most recent critical settings are protected during unplanned computer outages. This duplicate image can be stored either on an unused portion of the hard drive or on external devices, including CDs, DVDs, USB devices and networks. **Rockwell Automation; 800/223-5354, Ext. 1804; www. ra.rockwell.com**

HMI GETS WEB-ABILITY

Experion Station HMI uses a web-based architecture to provide a common infrastructure across the enterprise to ensure seamless, third-party



integration using open web standards. Its alarm and event presentation adheres to Abnormal Situation Management (ASM) Consortium and EEMUA Publication 191 guidelines. A Mobile Station version provides information to operators in the field. **Honeywell; 952/346-6409; www.honeywell.com**

ONE-WINDOW NAVIGATION

System 800xA Operations accesses enterprise-wide data from multiple applications or from any connected

workstation in the plant or office. It provides one window for navigating, accessing, and viewing plant information in real time. Contextual navigation presents the entire production facility in one window for single-point entry, single-point change, and re-use across the plant. **ABB; 203/750-2200; www.abb.com**

Resolutions Are Made to be Broken

Greg McMillan (gkmcmi@austin.rr.com) and Stan Weiner, PE (sweiner@swfla.rr.com) bring their wits and more than 66 years of process control experience to bear on your questions, comments, and problems.

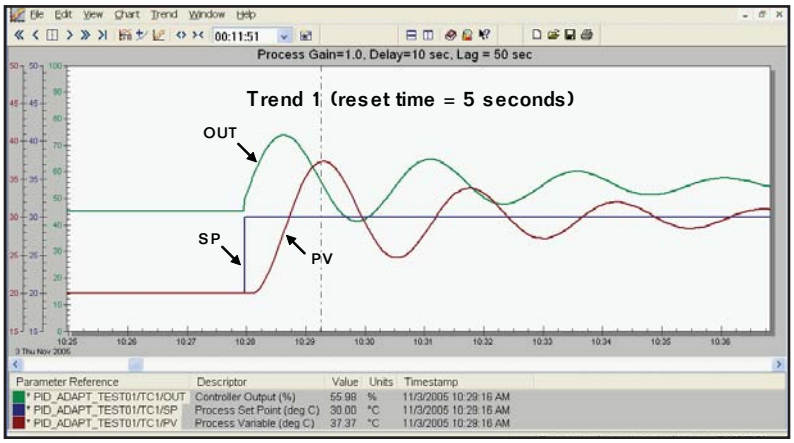
Stan In the first trend, which is the December puzzler (Figure 1), the process lag was increased from 5 seconds to 50 seconds.

Greg: The reset time that is correct for the dead-time dominant process is too short for the process where the process time constant (lag) was increased by a factor of 10. If the reset time is increased from 5 to 50 seconds, the slow oscillations and overshoot characteristic of too much reset action (too small a reset time) are eliminated. The only problem is that now the time to reach set point (rise time) is several times longer. This can be cut by one third by tripling the controller gain. This increase in controller gain is made possible by the time constant now being much slower than the dead time. In the second trend for the same process dynamics (Figure 2), the controller gain was increased from 0.25 to 0.75 and the reset time was increased from 5 to 50 seconds. Note that the controller output goes immediately to the final resting value.

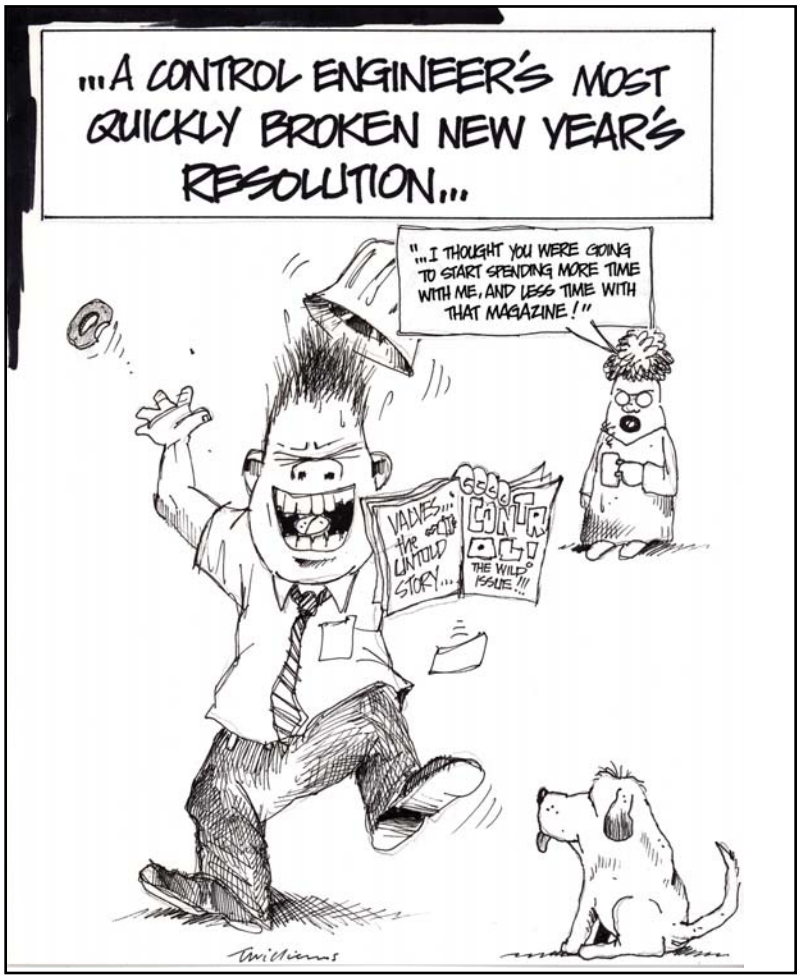
Stan: Finally, in the third trend (Figure 3), we changed a third tuning parameter. We added a rate setting of 150 seconds, which is about 10 times longer than it should be. The result is high-frequency oscillations in the controller output and a staircase in

FIGURE 1.

DON'T FALTER



Increasing the time constant from 5 to 50 seconds caused a slow reset cycle.



the set point response of the PV. It also is actually working against the set point change because the rate action is on PV. Even if rate on error, a set point filter and the proper amount of rate were used, it wouldn't improve the set point response much for this single-lag-plus-dead-time process.

Greg: In real life there is hardly ever just one lag, and rate can help if it is set equal to the second lag. For integrating processes, we can view the integrator gain as a big first lag, and set rate equal to the additional lag. The problem with derivative is that measurement noise is amplified. The best applications for rate are back-mixed temperature and pH loops.

Stan: We're going to stop using trend charts as puzzlers.

Greg: For you "Top 10 List" fans, we intend to keep the lists going since they are timeless.

"My Top 10 Broken New Year's Resolutions"

(10) Stop rooting against Texas teams now that my home is in Austin. Why did I enjoy the massacre of the Cowboys by the Redskins?

(9) Listen intently to my wife's instructions. Why does my mind still jump to weighty matters like what's next for dinner?

(8) Stop making cheap control valve jokes. Could the next final element reputation I hurt be my own?

(7) Help make smart diagnostics smarter. Do I need to de-fussify my fuzzy logic?

(6) Stop lusting in my heart for more computing power. Is it the PC or me that is the constraint?

(5) Turndown the volume on my headphones. What did you say?

(4) Stop drinking cheap wine. Does good wine ever come in a large enough-sized bottle?

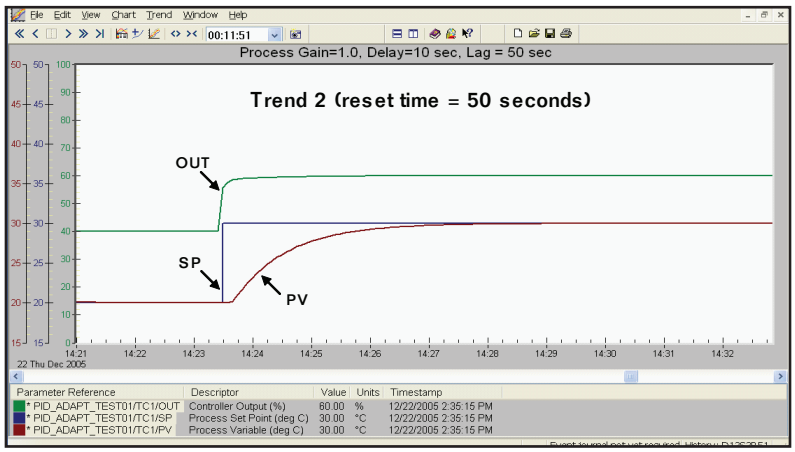
(3) Read a college text on control theory. Can I watch Star Trek without setting up the state space equations?

(2) Stop answering a question with a question. Why should a consultant do this?

(1) Spend more time with my wife than CONTROL magazine. Whatever happened to my January issue?

FIGURE 2.

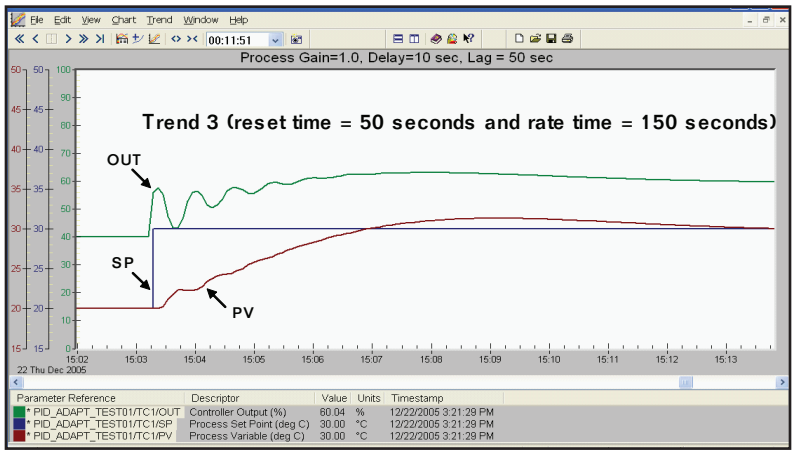
RESTING PLACE



In the second trend, increasing the controller gain from 0.25 to 0.75 and the reset time from 5 to 50 seconds causes the controller output to go to the final resting value, given the same process dynamics as the first trend.

FIGURE 3.

OBVIOUS OSCILLATIONS



Adding a rate setting of 150 seconds (which is about 10 times larger than it should be) in the third trend produces high-frequency oscillations in the controller output and a staircase in the set point response of the PV.

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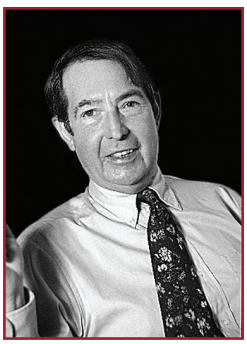
Advanced Process Control in Practice

Advanced process control is a concept that keeps evolving, always seeming just out of reach. In the beginning, it meant model-based process optimization under closed-loop regulatory control. This was something almost never realized in practice and seldom even attempted. Yet, in the 1960s, NASA was performing wonders with real-time programming and remote control that left industrial practitioners awe-struck with envy.

Two books published in 2004 by ISA (www.isa.org) and a new tool announced in the Fall of 2005 are heralding the advent of advanced process control as a standard tool for practicing engineers. The authors of these advances are all inductees in CONTROL magazine's Process Automation Hall of Fame, in addition to their distinguished business careers.



You really can't go wrong by listening to Wade, McMillan and Cutler.

Harold Wade's *Basic and Advanced Regulatory Control, 2nd Edition* is a thorough treatment of process control technology from basic PID through more sophisticated methods such as multivariable model predictive control. I've known Harold more than 25 years (see "The Process Software Alchemists," Around the Loop, CONTROL, Feb. '97, p. 90). In addition to being an accomplished industrial controls developer/implementer, Wade has written extensively, taught extensively, and designed PC ControLAB, an important process control training program. This book includes detailed discussions of controller tuning and self-tuning methods, cascade control, ratio control, feed-forward control and a host of specialized methods (split-range, cross-limiting, floating, time-proportioning and others).

Throughout his book, Wade's emphasis is on practical understanding of the concept rather than mathematical rigor in its derivation. This is not to imply that that his presentation is simplistic. The subject matter is complex and the mathematical tools must match this complexity. The objective, nonetheless, is to impart useful knowledge that can be applied by working professionals.

Models Unleashed by Gregory K. McMillan and Robert A. Cameron is a successor to *Advanced Control Unleashed* (2003), and to *Good Tuning* (2001). All three are pocket-sized books that aid working professionals by providing real-world operating problems to solve. In "Appendix D—Modern Myths" in *Models Unleashed*, the first myth is: "You need an advanced degree to do advanced control." The authors then proceed to demolish this myth.

Greg McMillan is currently on the faculty at Washington University in St. Louis and is a consultant through EDP Contract Services (Austin, Tex.). During his 33-year career with Monsanto and Solutia, he practiced process control in all of its many guises, and retired as an ISA senior fellow. McMillan also was recognized as 1994 Engineer of the Year by CONTROL and is a regular columnist with this publication. Robert Cameron was a close associate at Monsanto/Solutia, and has more than 18 years of high-level experience in control system programming and advanced control implementation.

Models Unleashed covers simulation models, model identification, model setup and model tuning. Its "Appendix E" is an excerpt from McMillan and Stan Weiner's, "Here Comes the Sludge," Control Talk, CONTROL, April '03, p. 73.

Finally, Cutler Technology Corp. announced Adaptive Dynamic Matrix Control (ADMC) for multivariable control on Oct. 1, 2005. A major advantage claimed for ADMC is its elimination of PID controllers from the control hierarchy. PID controller dynamics are removed, on-line identification of the process is facilitated, and deterioration of multivariable PID controllers with time is eliminated. ADMC's fundamental control algorithm is the field-proven DMC.

Charles R. Cutler developed and refined the DMC concept at Shell (Houston, Tex.), where he also was a colleague of Bill Biles. In 1984, Cutler organized DMCC as an independent software company. DMCC was acquired by Aspen Technology in the mid 1990s. Cutler Technology was organized in 1999 to advance DMC technology in cooperation with AspenTech.

You really can't go wrong by listening to Wade, McMillan and Cutler (in no particular order). All are eager to communicate the technology of advanced process control. More importantly, all are eager to have you translate this understanding into tangible applications.

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